INVERTIBILITY OF SWITCHED NONLINEAR SYSTEMS

BY

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THESIS

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ABSTRACT

Hybrid or switched systems can operate in several different modes, with some discrete dynamics governing the mode changes. Each mode of operation is described by a dynamical system having an internal state, an external input (which can be thought of as a disturbance or a control signal), and a measured output. An important structural property of these systems that has not been investigated is invertibility, which is the ability to reconstruct an unknown input and an unknown current mode of operation from the knowledge of the measured output and initial state. Invertibility is an important property in system design and system security analysis.

The thesis addresses the invertibility problem of switched systems where the subsystem dynamics are nonlinear but affine in controls. We extend the concept of switch-singular pairs, recently introduced by Vu and Liberzon, to nonlinear systems and develop a formula for checking if the given state and the output form a switch-singular pair. In case of single-input single-output systems, a simple rank test is provided to check for the existence of switch-singular pairs. We give a necessary and sufficient condition for a switched system to be invertible, which says that the subsystems should be invertible and there should be no switch-singular pairs. In case a switched system is invertible, one can build a switched inverse system to reconstruct the switching signal and the input. When all the subsystems are invertible, we present an algorithm for finding switching signals and inputs that generate a given output in a finite interval when there is a finite number of such switching signals and inputs. Detailed examples are included to illustrate these newly developed concepts. Two possible applications discussed in detail are fault detection in power systems and topology recovery in multiagent networks.

To my parents

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CHAPTER 1

INTRODUCTION

1.1 Background

Inversion of systems is an interesting problem from theoretical and practical viewpoints. The problem is, basically, concerned with finding the input from the given output and initial state. It has various applications in fields such as robotics [1, Chapter 12] and general control of mechanical systems [2], as well as in process control (see [3, Chapter 5] for examples related to electric circuits). Invertibility may represent a desired property in system design because it allows one to infer the system configuration (characterized by the operating mode) and the external influences (characterized by inputs) from the observed data. Or, to ensure system security, one may want to design the system not to be invertible, in order to ensure that the intruder will not be able to gain access to protected system information. From a control perspective, system inversion tools can also be used to find inputs that exactly track a desired output trajectory. For example, system inversion can be used to find control inputs that enable an aircraft to track a desired maneuver. In case of nonminimum phase systems, invertibility also plays a key role in stabilization and in output tracking problems with internal stability [4].

When dealing with switched systems, the systems under consideration consist of a family of dynamical subsystems together with a switching signal that determines the active subsystem at each time instant. Switching behaviors can come from controller design, such as in switching supervisory control [5] or gain scheduling control. Switching can also be inherent by nature, such as when a physical plant has the capability of undergoing several operational modes (e.g., an aircraft during different thrust modes [6, 7], a walking robot during leg impact and leg swing modes [8], different formations of a group of vehicles). Also, switched systems may be viewed as higher-level abstractions of hybrid systems obtained by neglecting the details of the discrete behavior and instead considering switching signals from a suitable class. As a result, switched systems have been a focus of ongoing research and several results related to stability, controllability, observability, and input-to-state stability of such systems have been published; see [5] for references. More recently, Vu and Liberzon introduced the problem of invertibility of switched linear systems in [9]. In the thesis, we extend their methodology to study the problem of invertibility of continuous-time switched nonlinear systems, which concerns with the following question: *What is the condition on the subsystems of a switched system such that, given an initial state x_0 and the corresponding output y generated with some switching signal σ and input u, we can recover the switching signal σ and the input u uniquely?* The problem statement is analogous to the classical invertibility problem for nonswitched systems. In fact, for every control system with an output, we have an input-output map and the question of left (respectively right) invertibility is, roughly speaking, that of the injectivity (respectively surjectivity) of this map.

1.2 Previous Work

System invertibility problems are of great importance from theoretical and practical viewpoints and have been studied extensively for 50 years, after being pioneered by Brockett-Mesarovic [10]. The systematic study of invertibility for nonswitched nonlinear systems began with Hirschorn, who first studied the single-input single-output (SISO) case [11] and then generalized Silverman's structure algorithm for multiple-input multiple-output (MIMO) nonlinear systems [12]. Singh [13] then modified the algorithm to cover a larger class of systems. Isidori and Moog [14] used this algorithm to calculate zero-output constrained dynamics and reduced inverse system dynamics. The algorithm is also closely related to the dynamic extension algorithm used to solve the dynamic state feedback input-output decoupling problem [15, Sections 8.2 and 11.3]. The geometric methods have also been used by Nijmeijer [16]. A higher-level interpretation given by a linear-algebraic framework, which also establishes links between these algorithms and the geometric approach, is presented by Di Benedetto et al. in [17]. We also recommend [18, Chapter 5] for a useful survey on various invertibility techniques.

The problem of invertibility for switched linear systems was introduced very recently in [9], where the authors used Silverman's structure algorithm to formulate conditions for the invertibility of switched systems with continuous dynamics. The problem of invertibility for discrete time switched linear systems has been discussed in [19, 20], but here the authors assume that the switching sequence is known and find the corresponding input. This thesis addresses the invertibility of switched nonlinear systems with unknown inputs and unknown switching signals,[1] a problem which has many interesting potential applications but has not been explored thus far.

1.3 Thesis Contribution

In our research work, we adopt an approach similar to [9] and use Singh's nonlinear structure algorithm to formulate conditions for the invertibility of switched

[1]A related problem is discussed in [21] but it does not follow the same theoretical approach we do, and instead uses a heuristic approach with the purpose of studying a specific application.

systems with continuous-time nonlinear dynamical subsystems that are affine in controls. Although the form of the main condition (invertibility of subsystems plus no switch-singular pairs) is essentially similar to [9], the technical details of checking the conditions are different because we work with the nonlinear structure algorithm.

As is the case in the classical setting with nonswitched systems, we start with an output and an initial state, but here there is a set of dynamic models and we wish to recover the switching signal in addition to the input. In the context of hybrid systems, recovering the switching signal is equivalent to *mode identification* for hybrid systems, which has been studied in [22, 23]. Hence, for the inversion of switched systems, we wish to do *mode detection* and *input recovery* at the same time. Consequently, the basic idea for solving the invertibility problem is to do mode identification by utilizing the relationship among the outputs and the states of the subsystems and then using the nonlinear structure algorithm for the corresponding subsystem to recover the input. For the case when subsystems are linear, Silverman's structure algorithm seems to be the most convenient tool to formulate invertibility conditions which lead to simple and elegant rank test for checking the existence of switch-singular pairs, but in nonlinear systems it is hard to achieve such a level of generality. For this reason, we start with the SISO case to highlight the technical difficulties in moving from linear to nonlinear systems. Discussing the SISO case first also helps in understanding the concepts behind the formula derived for verification of switch-singular pairs. These formulae reduce to nice "rank tests" when the subsystem dynamics are SISO and bilinear. For MIMO systems, Hirschorn's algorithm, although simpler, covers a relatively limited class of systems as compared to the one given by Singh. So, we use Singh's modified structure algorithm to develop the relationship between the output, the input, and the state, but the conditions for invertibility get more involved as compared to linear systems.

The tool set developed for checking invertibility of switched systems has also been used for the output tracking, where we find inputs and switching signals that can generate a given output from a given initial state even if the switched system is not invertible. We provide a switching inversion algorithm that produces all inputs and switching signals capable of generating a given function in a finite time interval from a given initial state, when there are finitely many such inputs and switching signals. In the algorithm, the concept of switch-singular pair plays a crucial role as it allows us to identify potential switching times even though the output may be smooth at these times.

1.4 Thesis Organization

The thesis is organized as follows. Chapter 2 contains the definitions of invertibility and the formal problem statement followed by the main result on left invertibility. We then give a characterization of switch-singular pairs and the construction of inverse systems in Chapter 3. Output tracking is discussed in Chapter 4, which also contains the algorithm for output generation along with a detailed example. We discuss two possible applications of our theoretical work in Chapter 5. We conclude the thesis with some remarks on further research directions.

CHAPTER 2

CHARACTERIZATION OF INVERTIBILITY

2.1 Preliminaries

In this section, we develop the required notations and provide some background on invertibility of nonswitched nonlinear systems. Based on that, we develop the definition for the invertibility of switched nonlinear systems followed by the formal problem statement to which we seek solution in the thesis.

2.1.1 Nonswitched nonlinear systems

The dynamics of a square nonlinear system, affine in controls, are given by

$$\Gamma := \begin{cases} \dot{x} = f(x) + G(x)u = f(x) + \sum_{i=1}^{m} g_i(x)u_i, \\ y = h(x) \end{cases} \tag{2.1}$$

where $x \in \mathbb{M}$, an n-dimensional real connected smooth manifold, for example $\mathbb{R}^n$; f, g_i are smooth vector fields on $\mathbb{M}$, and $h : \mathbb{M} \to \mathbb{R}^m$ is a smooth function.

We start off by reviewing classical definitions of invertibility for such systems. For that, consider the input-output map $\mathrm{H}_{x_0} : \mathcal{U} \to \mathcal{Y}$ for some input function space $\mathcal{U}$ and the corresponding output function space $\mathcal{Y}$. Since the state trajectories of nonlinear systems exhibit finite blow-up times, some input signals may not have a well defined image in the output space, over the same length of interval, under this map. We do not give a rigorous definition of H_{x_0} but use it, nevertheless, for better illustration. It is assumed that the outputs exist on the intervals considered. The input space $\mathcal{U}$ is not specified at this moment as it depends upon the system under consideration (see Chapter 3). Denote by $\Gamma_{x_0}(u)$ the trajectory of the corresponding system with the initial state x_0 and the input u, and the corresponding output by $\Gamma^O_{x_0}(u)$. If η is a function of time, then $\eta_{[t_0,T]}$ denotes that function with domain restricted to $[t_0, T]$. In the case of nonswitched systems, the following notion of invertibility[1] was introduced in [12].

Definition 1 Given any output $y \in \mathcal{Y}$ on an arbitrary interval $[t_0, T]$, the system (2.1) is *invertible at a point* $x_0 := x(t_0) \in \mathbb{M}$ if $\Gamma^O_{x_0}(u_{1[t_0,T]}) = \Gamma^O_{x_0}(u_{2[t_0,T]}) = y_{[t_0,T]}$ implies that $\exists\, \varepsilon > 0$ such that $u_{1[t_0,t_0+\varepsilon)} = u_{2[t_0,t_0+\varepsilon)}$. The system is *strongly invertible at a point* x_0 if it is invertible for each $x \in N(x_0)$, where N is some open neighborhood of x_0. The system is *strongly invertible* if there exists an open and dense submanifold $\mathbb{M}^\alpha$ such that $\forall x_0 \in \mathbb{M}^\alpha$, the system is strongly invertible at x_0. $\triangleleft$

By definition, it can be inferred that invertibility is equivalent to the injectivity of H_{x_0}; that is, $\mathrm{H}_{x_0}(u_{1[t_0,t_0+\varepsilon)}) = \mathrm{H}_{x_0}(u_{2[t_0,t_0+\varepsilon)}) \Leftrightarrow u_{1[t_0,t_0+\varepsilon)} = u_{2[t_0,t_0+\varepsilon)}$.

In general, the inverses of nonlinear dynamical systems are not defined globally. An open and dense subset of $\mathbb{M}$ on which the dynamics of a nonlinear system are invertible is called the *inverse submanifold* and is denoted by $\mathbb{M}^\alpha$. If $x \in \mathbb{M}\backslash\mathbb{M}^\alpha$, we call it a *singular point* as it is not possible to invert the system starting from such an initial condition. In the most general construction of inverse systems, as the one given by Singh [13], there exists a set of *singular outputs* $\mathcal{Y}^s$ such that the system is not invertible for $y \in \mathcal{Y}^s$; and its complement $\mathcal{Y}^\alpha := \mathcal{Y}\backslash\mathcal{Y}^s$ is the set of all outputs for which the system is strongly invertible. All these notions will be

[1]Throughout the thesis, *invertibility* refers to left invertibility.

developed formally in Chapter 3, but here we give an example to show how these notions appear naturally when dealing with nonlinear systems.

Example 1 Consider a nonswitched nonlinear system with two inputs and outputs,

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} x_1 u_1 \\ x_3 u_1 \\ u_2 \end{pmatrix}, \quad \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mathbb{M} = \mathbb{R}^3.$$

We then have

$$\dot{y}_1 = x_1 u_1, \tag{2.2a}$$

$$\ddot{y}_2 = \frac{x_3 \ddot{y}_1 - \dot{y}_1 \dot{y}_2 + \dot{y}_1 u_2}{x_1}. \tag{2.2b}$$

So, y_1 is in one-to-one correspondence with u_1 if $x_1 \neq 0$, and y_2 is in one-to-one relation with u_2 if $\dot{y}_1 \neq 0$. Consequently, the set $\mathbb{M}^\alpha = \{x \in \mathbb{R}^3 \,|\, x_1 \neq 0\}$ is the inverse submanifold and the set of singular outputs is $\mathcal{Y}^s = \{y(t) \in \mathbb{R}^2 \,|\, \dot{y}_1(t) = 0\}$. If $x(t) \in \mathbb{M}^\alpha$ and $y(t) \notin \mathcal{Y}^s$ for all $t \in [t_0, t_0 + \varepsilon)$, then there is a one-to-one relation between the output and input signals provided their domains are restricted to $[t_0, t_0 + \varepsilon)$. By using Definition 1, we deduce that the system is strongly invertible. $\triangleleft$

From Example 1, it is clear that the system (2.1) is invertible at every $x \in \mathbb{M}^\alpha$ for the class of inputs u such that along the trajectory of the system (2.1), the resulting motion $(x(t), y(t)) \in \mathbb{M}^\alpha \times \mathcal{Y}^\alpha$. Hence, invertibility is achieved when the domain of signals is restricted to $[t_0, T)$ with $T \in (t_0, \bar{t})$ and $\bar{t} := \min\{t > t_0 : (x(t), y(t)) \notin \mathbb{M}^\alpha \times \mathcal{Y}^\alpha\}$. In other words, a nonlinear system is invertible at x_0 if for a given output y over the time interval $[t_0, T']$, one can find $T \in (t_0, T']$ and a unique input u over $[t_0, T)$ such that $\Gamma^O_{x_0}(u_{[t_0,T)}) = y_{[t_0,T)}$. Since T can be arbitrarily small, this explains why we require distinct inputs over arbitrarily small time domains in Definition 1. We will generalize this notion of local invertibility to the switched systems.

2.1.2 Switched nonlinear systems

A finite family of systems defined by (2.1) generates a switched system and in the thesis we will consider such switched nonlinear systems, affine in controls, that have the following structure:

$$\Gamma_\sigma : \begin{cases} \dot{x} = f_\sigma(x) + G_\sigma(x)u = f_\sigma(x) + \sum_{i=1}^{m}(g_i)_\sigma(x)u_i, \\ y = h_\sigma(x) \end{cases} \tag{2.3}$$

where $\sigma : [0, \infty) \to \mathcal{P}$ is the switching signal that indicates the active subsystem at every time, $\mathcal{P}$ is some finite index set, and f_p, G_p, h_p, where $p \in \mathcal{P}$, define the dynamics of individual subsystems. The state space $\mathbb{M}$ is a connected real smooth manifold of dimension n, for example $\mathbb{R}^n$; f_p, $(g_i)_p$ are real smooth vector fields on $\mathbb{M}$, and $h_p : \mathbb{M} \to \mathbb{R}^m$ is a smooth function. A switching signal, as defined in [5], is a piecewise constant and everywhere right-continuous function that has a finite number of discontinuities τ_i, which we call *switching times*, on every bounded time interval and thus $\sigma(t) = p \in \mathcal{P}$, $\forall\, t \in [\tau_i, \tau_{i+1})$. Denote by σ^p the constant switching signal such that $\sigma^p(t) = p$, $\forall\ t > 0$. We assume that all the subsystems are equidimensional, they live in the same state space $\mathbb{M}$, and that there is no state jump at switching times. For any initial state x_0, switching signal σ, and piecewise continuous input u on any time domain, a solution of (2.3) over the same domain always exists (in Carathéodory sense) and is unique, provided the flow of the active subsystem is defined $\forall\, t \in [\tau_i, \tau_{i+1})$. In case of no switching, this condition is equivalent to the forward completeness of the flow and we assume that each subsystem satisfies this condition. For $p \in \mathcal{P}$, denote by $\Gamma_{p,x_0}(u)$ the trajectory of the corresponding subsystem with the initial state x_0 and the input u, and the corresponding output by $\Gamma^O_{p,x_0}(u)$. Since switching signals are right-continuous, the outputs are also right-continuous and whenever we take derivative of the output, we assume it is the right derivative.

We will use $\mathcal{F}^{pc}$ to denote the space of *piecewise right-continuous functions*[2] and $\mathcal{F}^n$ to denote the subset of $\mathcal{F}^{pc}$ whose elements are n times differentiable between two consecutive discontinuities. Likewise, $\mathcal{F}^{AC}$ denotes the subset of $\mathcal{F}^{pc}$ whose elements are absolutely continuous (defined in [24]) between two consecutive discontinuities. Throughout the document, we will regularly use the concatenations of functions and we formally define the concatenation map as follows: let $y_i \in \mathcal{F}^{pc}_{[t_i,\tau_i)}$, $i = \{1, 2\}$; define the *concatenation map* $\oplus : \mathcal{F}^{pc} \times \mathcal{F}^{pc} \to \mathcal{F}^{pc}$ as

$$(y_1 \oplus y_2)(t) = \begin{cases} y_1(t) & \text{if } t \in [t_1, \tau_1), \\ y_2(t_2 + t - \tau_1) & \text{if } t \in [\tau_1, \tau_1 + \tau_2 - t_2). \end{cases}$$

Basically, $y_1 \oplus y_2$ means putting the two functions together in that order. Note that $y_1 \oplus y_2 = y_1 \forall\, y_2$ if $\tau_1 = \infty$, and in general, $y_1 \oplus y_2 \neq y_2 \oplus y_1$. The concatenation of vector functions is defined as the pairwise concatenation of the corresponding elements. The concatenation of an element y and a set S is $y \oplus S := \{y \oplus \zeta, \zeta \in S\}$. By convention, $y \oplus \emptyset = \emptyset$, $\forall\, y$, where $\emptyset$ is an empty set. Finally, the concatenation of two sets S and T is $S \oplus T := \{\eta \oplus \zeta,\, \eta \in S,\, \zeta \in T\}$.

In case of switched systems (2.3), the map H_{x_0} has an augmented domain; that is,

[2]By piecewise right-continuous functions, we mean that there is a finite number of jump discontinuities in any finite interval; there is no hole discontinuity and vertical asymptote; the function is continuous in between any two consecutive discontinuities; and the function is continuous from the right at discontinuities. To avoid excessive rigidness, we will use the term "piecewise continuous" throughout the paper, and it is understood that "piecewise continuous" means "piecewise right-continuous."

now we have a (switching signal $\times$ input)-output map $\mathrm{H}_{x_0} : \mathcal{S} \times \mathcal{U} \to \mathcal{Y}$, where $\mathcal{S}$ is a switching signal set. Let us first extend the definition of invertibility of nonswitched systems to define the invertibility of the map H_{x_0} for switched systems.

Definition 2 Given any output $y \in \mathcal{Y}$ on an arbitrary interval $[t_0, T]$, a *switched system is invertible at a point* x_0 if $\mathrm{H}_{x_0}(\sigma_{1[t_0,T]}, u_{1[t_0,T]}) = \mathrm{H}_{x_0}(\sigma_{2[t_0,T]}, u_{2[t_0,T]}) = y_{[t_0,T]}$ implies that $\exists\ \varepsilon > 0$ such that $\sigma_{1[t_0,t_0+\varepsilon)} = \sigma_{2[t_0,t_0+\varepsilon)}$ and $u_{1[t_0,t_0+\varepsilon)} = u_{2[t_0,t_0+\varepsilon)}$. A *switched system is strongly invertible at a point* x_0 if it is invertible at each $x \in N(x_0)$, where N is some open neighborhood of x_0. A *switched system is strongly invertible* if there exists an open and dense submanifold $\mathbb{M}^\alpha$ of $\mathbb{M}$ such that $\forall x_0 \in \mathbb{M}^\alpha$, the system is strongly invertible at x_0 for given $y \in \mathcal{Y}$. $\lhd$

As was the case in nonswitched systems, the invertibility of switched systems requires the preimage of H_{x_0} to be unique on some interval for given x_0 and y. The reason we have a different notion of invertibility is because in switched systems, if a subsystem is invertible at x_0 for a given nonsingular output y, then it is possible that another subsystem might produce the same output starting from the same initial condition. This means that the pre-image of H_{x_0} at such (x_0, y) is not unique and hence the switched system is not invertible at x_0 if such pairs (x_0, y) exist. We call all such pairs *switch-singular pairs*.[3] The concept of switch-singular pairs for switched systems basically refers to the ability of more than one subsystem to produce a segment of the desired output starting from the same initial condition. The formal definition is given below:

Definition 3 Let $x_0 \in \mathbb{M}$ and $y \in \mathcal{Y}$ on some time interval $[t_0, T]$. The pair (x_0, y) is a *switch-singular pair* of the two subsystems Γ_p, Γ_q if there exist u_1, u_2 and $\varepsilon > 0$ such that $\Gamma^O_{p,x_0}(u_{1[t_0,t_0+\varepsilon)}) = \Gamma^O_{q,x_0}(u_{2[t_0,t_0+\varepsilon)}) = y_{[t_0,t_0+\varepsilon)}$. $\lhd$

[3]This is similar to the concept of singular pairs conceived in [9]. We use the term "switch-singular pair" to avoid conflict with the singularities of individual nonlinear subsystems.

If all subsystems are linear, $x_0 = 0$ and $y \equiv 0$ always form a switch-singular pair regardless of the dynamics of the subsystems. This is because $u \equiv 0$ and any switching signal will produce $y \equiv 0$, that is, $\mathrm{H}_0(\sigma, 0) = 0 \ \forall \ \sigma$, and therefore H_0 is not injective if the function $0 \in \mathcal{Y}$. In nonlinear systems, this is not the case in general and all switch-singular pairs are solely determined by the subsystem dynamics. As stated earlier, the switched system is not invertible if $\mathcal{Y}$ contains the outputs that form switch-singular pairs. Thus, if there exist any switch-singular pairs, we have to restrict the output set $\mathcal{Y}$, instead of letting $\mathcal{Y}$ to be the set of all possible concatenations of nonsingular output trajectories.

The discussion motivates us to formally define the invertibility problem for switched nonlinear systems.

The invertibility problem: Consider a (switching signal $\times$ input)-output map $\mathrm{H}_{x_0} : \mathcal{S} \times \mathcal{U} \to \mathcal{Y}$ for the switched system (2.3). Find the largest possible set $\mathcal{Y}$, an open dense set in $\mathbb{M}$ and a condition on the subsystems such that for a given output $y \in \mathcal{Y}$ over a finite time interval $[t_0, T']$, there exist $T \in (t_0, T']$ and a unique (σ, u) over $[t_0, T)$ having the property that $\mathrm{H}_{x_0}(\sigma_{[t_0,T)}, u_{[t_0,T)}) = y_{[t_0,T)}$.

2.2 Solution to the Invertibility Problem

We now give conditions on the subsystem dynamics so that the map H_{x_0} is injective for some sets $\mathcal{S}, \mathcal{U}$, and $\mathcal{Y}$. We do not explicitly specify what the input sets $\mathcal{U}$ and $\mathcal{S}$ are, but instead we specify the set $\mathcal{Y}$ and then $\mathcal{U}$ will be the corresponding set that, together with $\mathcal{S}$, generates $\mathcal{Y}$.

For all $p \in \mathcal{P}$, let $\mathbb{M}_p^\alpha$ be the inverse submanifold of Γ_p, $\mathcal{Y}_p$ be the set of smooth[4] outputs that can be generated by Γ_p, $\mathcal{Y}_p^s$ be the set of singular outputs of Γ_p, and

[4]This assumption can be relaxed depending upon the system under consideration as discussed in Chapter 3.

$\mathcal{Y}_p^\alpha$ be the set of outputs on which Γ_p is strongly invertible. Define $\mathcal{Y}^s := \cup_{p\in\mathcal{P}}\mathcal{Y}_p^s$ as the collection of all singular outputs and let $\mathcal{Y}^{all}$ be the set of outputs generated by all the possible concatenations of all elements of $\mathcal{Y}_p$, $\forall p \in \mathcal{P}$. Then $\mathcal{Y}^\alpha := \mathcal{Y}^{all}\backslash\mathcal{Y}^s$ is a set of outputs on which every subsystem is strongly invertible. We consider outputs $y \in \mathcal{Y}^\alpha$ over a finite interval $[t_0,\, T']$ and seek invertibility on a subinterval $[t_0,\, T) \subset [t_0,\, T']$ such that $(\sigma_{[t_0,\,T)}, u_{[t_0,\,T)})$ is a unique pre-image of $y_{[t_0,\,T)}$.

The first main result is about the strong invertibility at some $x_0 \in \mathbb{M}$.

Theorem 1 *Consider the switched system (2.3) and the output set $\mathcal{Y}^\alpha$. The switched system is strongly invertible at $x_0 \in \mathbb{M}$ for given $y \in \mathcal{Y}^\alpha$ if and only if there exists a neighborhood $N(x_0)$ such that each subsystem is invertible at every $x \in N(x_0)$, and for all $x \in N(x_0)$, $y \in \mathcal{Y}^\alpha$, the pairs (x, y) are not switch-singular pairs of Γ_p, Γ_q for all $p \neq q$, and $p, q \in \mathcal{P}$.*

Proof. *Necessity*: We show that if any of the subsystems is not strongly invertible at x_0 or if there exist switch-singular pairs, then the switched system is not invertible.

Suppose that a subsystem Γ_p, $p \in \mathcal{P}$, is not invertible at some x in arbitrary $N(x_0)$, then $\Gamma_{p,x}$ is not injective. Pick arbitrary $y \in \mathcal{Y}^\alpha \cap \mathcal{Y}_p^\alpha$, which is always possible by the definition of $\mathcal{Y}^\alpha$. Since $y \in \mathcal{Y}_p^\alpha$, and Γ_p is not invertible at x, it follows that for every $\varepsilon > 0$, $\exists\, u_1 \neq u_2$ over time interval $[t_0, t_0+\varepsilon) \subset [t_0, T']$ such that $\Gamma^O_{p,x}(u_1) = \Gamma^O_{p,x}(u_2) = y_{[t_0,t_0+\varepsilon)}$. This implies that $\mathrm{H}_x(\sigma^p, u_1) = \mathrm{H}_x(\sigma^p, u_2) = y$, and the map H_x is not injective for given y. Hence, the switched system is not invertible at x. Since there exists such x in every neighborhood of x_0, it follows that the switched system is not strongly invertible at x_0.

For necessity of the second condition, suppose that $\exists\, x \in N(x_0)$, $y \in \mathcal{Y}^\alpha \cap \mathcal{C}^\infty$ such that $(x,\, y)$ is a switch-singular pair of Γ_p, Γ_q, $p \neq q$. This means that both subsystems, even though invertible at x, can produce this output over the interval $[t_0, t_0 + \varepsilon) \subset [t_0, T']$, $\forall \varepsilon > 0$. Consequently, $\exists\ u_1,\ u_2$ (possibly same) on the corresponding interval such that $\Gamma^O_{p,x}(u_1) = \Gamma^O_{q,x}(u_2) = y$. Hence, we have $\mathrm{H}_x(\sigma^p, u_1) = \mathrm{H}_x(\sigma^q, u_2) = y$; that is, the preimage of y is not unique as $\sigma^p \neq \sigma^q$. This implies that the switched system is not invertible at x for given $y \in \mathcal{Y}^\alpha$. Since there exists such x in every neighborhood of x_0, it follows that the switched system is not strongly invertible at x_0.

Sufficiency: Suppose that for given $x_0 \in \mathbb{M}$, there exist some inputs u_1, u_2 and switching signals σ_1, σ_2 such that $\mathrm{H}_{x_0}(\sigma_1,\ u_1) = \mathrm{H}_{x_0}(\sigma_2,\ u_2) = y \in \mathcal{Y}^\alpha$ over $[t_0, T']$. Initially, we have $\sigma_1(t_0) = \sigma_2(t_0) = p$ because $(x_0,\ y)$ is not a switch-singular pair. Since $y \in \mathcal{Y}^\alpha_p$, and Γ_p is invertible at every $x \in N(x_0)$, $\exists\ \varepsilon_1 > 0$ such that $u_{1[t_0,t_0+\varepsilon_1)} = u_{2[t_0,t_0+\varepsilon_1)} = \Gamma^{-1,O}_{p,x}(y_{[t_0,t_0+\varepsilon_1)})$, the output of the inverse subsystem. As there are no switch-singular pairs in $N(x_0)$, $\exists\ \varepsilon_2 > 0$ such that $\sigma_{1[t_0,t_0+\varepsilon_2)} = \sigma_{2[t_0,t_0+\varepsilon_2)}$. Letting $\varepsilon = \min\{\varepsilon_1, \varepsilon_2\}$, it then follows from Definition 2 that the switched system is invertible at every $x \in N(x_0)$ and hence is strongly invertible at x_0. $\square$

To get some intuition behind the result, note that for any output y that is generated by the switched system, at any time instant t, there exists $p \in \mathcal{P}$ such that $y(t) \in \mathcal{Y}^\alpha \cap \mathcal{Y}^\alpha_p$. Since each subsystem Γ_p is invertible, there exists a unique input which produces that output. Nonexistence of switch-singular pairs implies that no other system can produce the same output even with different input. Hence, $\mathrm{H}^{-1}_{x_0}(y) = (\sigma,\ u)$ is unique.

Based on the result of Theorem 1, the conditions for strong invertibility of switched systems can be developed. Let $\mathbb{M}^\alpha := \bigcap_{p\in\mathcal{P}} \mathbb{M}^\alpha_p$, then $\mathbb{M}^\alpha$ is an open and dense subset of $\mathbb{M}$ because it is a finite intersection of open and dense subsets. Since every subsystem is strongly invertible on $\mathbb{M}^\alpha$, we have the following result.

Corollary 1 *The switched system (2.3) is strongly invertible at every $x_0 \in \mathbb{M}^\alpha$ and for $y \in \mathcal{Y}^\alpha$ if and only if Γ_p, $\forall\, p \in \mathcal{P}$, is strongly invertible at every $x_0 \in \mathbb{M}^\alpha_p$ and* *the subsystem dynamics are such that the pairs* (x_0, y) *are not switch-singular pairs of* Γ_p, Γ_q *for all* $p \neq q$, $p, q \in \mathcal{P}$, *for every* $x_0 \in \mathbb{M}^\alpha$, *and every* $y \in \mathcal{Y}^\alpha$. $\triangleleft$

It follows from the proof of the sufficiency part in Theorem 1 that the switched system is strongly invertible over the interval $[t_0, T)$, where $T \in [t_0, \bar{t})$ and $\bar{t} := \min\{t > t_0 : (x(t), y(t)) \notin \mathbb{M}^\alpha \times \mathcal{Y}^\alpha\}$. If the output y loses continuity over the interval $[t_0, T)$ because of switching, then $(\sigma_{[t_0,T)}, u_{[t_0,T)}) = (\sigma_{[t_0,\tau_1)}, u_{[t_0,\tau_1)}) \oplus \cdots \oplus (\sigma_{[\tau_k,T)}, u_{[\tau_k,T)})$, where k is the total number of switches in the interval $[t_0, T)$ and τ_i, $i = 1, \cdots, k$, are the switching instants.

Remark 1 *For the switched system (2.3), if all the subsystems are globally invertible in addition to the hypothesis of Corollary 1, that is,* $\mathbb{M}^\alpha = \mathbb{M}$ *and* $\mathcal{Y}^s = \emptyset$, *then the domain of signals can be arbitrary such as* $[t_0, \infty)$ *and the switched system is strongly invertible for outputs that are defined over* $[t_0, T)$, $\forall\, T > t_0$. $\triangleleft$

Remark 2 *If a subsystem has more inputs than outputs, then it cannot be (left) invertible. On the other hand, if it has more outputs than inputs, then some outputs are redundant (as far as the task of recovering the input is concerned). Thus, the case of input and output dimensions being equal is, perhaps, the most interesting case.* $\triangleleft$

Theorem 1 characterizes the invertibility of switched nonlinear systems in an abstract setting. It is still unclear how one would proceed mathematically to verify the required hypothesis. We discuss this issue in next chapter.

CHAPTER 3

CHECKING INVERTIBILITY

This chapter addresses the computational aspect of the concepts introduced in the previous chapter and leads to the development of algebraic criteria for checking the invertibility of switched systems. The first condition in Theorem 1 asks for invertibility of subsystems and is verified by the structure algorithm. To put everything into perspective, we provide appropriate background related to the invertibility of nonswitched systems and use it to develop the concept of functional reproducibility. To check if (x_0, y) is a switch-singular pair, we develop a formula using the functional reproducibility criteria of nonswitched systems. Based on these two mathematical characterizations and the result in Theorem 1, we will be able to construct a switched inverse system for recovering the original input and switching signal uniquely.

3.1 Single-Input Single-Output (SISO) Systems

We start off with the case when all the subsystems are SISO because it gives more insight into computations and helps understand the concepts which we will later generalize to multivariable systems. To this end, consider a SISO nonlinear system affine in controls,

$$\begin{aligned} \dot{x} &= f(x) + g(x)u, \qquad (3.1) \\ y &= h(x) \end{aligned}$$

where x evolves on a manifold $\mathbb{M}$, $f(x)$ and $g(x)$ are smooth vector fields, and $h(x)$ is a smooth function from $\mathbb{M}$ to $\mathbb{R}$. Assume that the system (3.1) has a relative degree r at x_0 [25], i.e., $\exists$ a neighborhood $N(x_0)$ such that $L_g L_f^{r-1} h(x) \neq 0 \; \forall \, x \in N(x_0)$, where $L_f^k h(x) = \frac{\partial(L_f^{k-1} h(x))}{\partial x} f(x)$ and $L_f^0 h(x) = h(x)$. To check if this nonswitched system is invertible or not, we first derive an explicit expression for the input u in terms of the output y by computing the derivatives of y as follows:

$$\begin{aligned} y(t) &= h(x(t)) && (3.2a) \\ \dot{y}(t) &= L_f h(x(t)) && (3.2b) \\ &\vdots \\ y^{(r)}(t) &= L_f^r h(x) + L_g L_f^{r-1} h(x) u(t). && (3.2c) \end{aligned}$$

From the last equation, we can derive an expression for $u(t)$:

$$u(t) = -\frac{L_f^r h(x)}{L_g L_f^{r-1} h(x)} + \frac{1}{L_g L_f^{r-1} h(x)} y^{(r)}(t). \tag{3.3}$$

Hence, u can be determined explicitly in terms of measured output y. On substituting the expression for u from (3.3) in Equation (3.1), one gets the dynamics for the inverse system:

$$\begin{aligned} \dot{z} &= f(z) + g(z)\Big(-\frac{L_f^r h(z)}{L_g L_f^{r-1} h(z)} + \frac{1}{L_g L_f^{r-1} h(z)} y^{(r)}(t) \Big), \\ u(t) &= -\frac{L_f^r h(z)}{L_g L_f^{r-1} h(z)} + \frac{1}{L_g L_f^{r-1} h(z)} y^{(r)}(t) \end{aligned} \tag{3.4}$$

where z is the state of the inverse system which evolves continuously with time, and for the sake of a compact expression, we have suppressed its time argument. The dynamics of this inverse subsystem evolve on the set $\mathbb{M}^\alpha := \{z \in \mathbb{M} \mid L_g L_f^{r-1} h(z) \neq$

$0\}$. Since the inverse system dynamics are driven by $y^{(r)}(t)$ which satisfies Equation (3.2c), it is not hard to see that the state trajectories of the inverse system satisfy the differential equation of the original system (3.1). So, if the inverse system is initialized with the same initial condition as that of the plant, then both of the systems follow exactly the same trajectory. The discussion motivates the following result, given in [11]:

Lemma 1 *A SISO system is strongly invertible at x_0 if and only if the system has a finite relative degree r at x_0.*

We developed the proof of the sufficiency part. The necessity part, although intuitively clear, is proved rigorously in [11].

For SISO systems, the input u appears in the r-th derivative of the output (3.2). Thus, the differentiability/smoothness of u will not affect the existence of the first $r-1$ derivatives of y. If $u : [0, t) \to \mathbb{R}$ is a locally essentially bounded, Lebesgue measurable function, then $y^{(r)}(t)$ exists almost everywhere and $y^{(r-1)}(t)$ is absolutely continuous [24]. So for SISO nonlinear nonswitched systems, $\mathcal{U}$ can be defined as the space of functions which are locally essentially bounded and Lebesgue measurable; and $\mathcal{Y}$ can be the set of corresponding outputs.

We now turn to the concept of functional reproducibility, which in broad terms means the ability to follow a given reference signal. This concept will help us study the existence of switch-singular pairs. We look at the conditions under which a system can produce the desired output y_d over some interval $[t_0, T)$ starting from a particular initial state x_0. To be precise, given the system (3.1) and an initial state x_0, we want to find out if there exists a control u such that $\Gamma^O_{x_0}(u) = y_d(\cdot)$. The following result was given in [11]:

Lemma 2 *If the system (3.1), with $x(t_0) = x_0$, has a relative degree $r < \infty$ at x_0,* *then there exists a control input u such that $\Gamma^O_{x_0}(u(t)) = y_d(t)$, for all $t \in [t_0, t_0+\varepsilon)$, and $\varepsilon > 0$, if and only if*

$$y_d^{(k)}(t_0) = L_f^k h(x_0) \qquad \forall\, k = 0, 1, \cdots, r-1 \tag{3.5}$$

Proof *Necessity:* Supposing $\exists\ \varepsilon > 0$ and input u such that $\Gamma^O_{x_0}(u(t)) = y_d(t)$, $\forall\ t \in [t_0, t_0+\varepsilon)$, then

$$\begin{aligned} y_d(t_0) &= y(t_0) = h(x_0) \\ \dot{y}_d(t_0) &= \dot{y}(t_0) = L_f h(x_0) \\ &\vdots \\ y_d^{(r-1)}(t_0) &= y^{(r-1)}(t_0) = L_f^{r-1} h(x_0). \end{aligned} \tag{3.6}$$

Sufficiency: If we inject $y_d(t)$ into the inverse system (3.4), then the control input produced by this inverse system is given by

$$u(t) = -\frac{L_f^{(r)} h(x)}{L_g L_f^{r-1} h(x)} + \frac{1}{L_g L_f^{r-1} h(x)} y_d^{(r)}(t). \tag{3.7}$$

Since $L_g L_f^{r-1} h(x_0) \neq 0$, u is well defined on an interval $[t_0, t_0+\varepsilon)$, for some $\varepsilon > 0$. When system (3.1) is driven by this input, then using Equation (3.2), we have

$$y^{(r)}(t) = L_f^{(r)} h(x(t)) + y_d^{(r)}(t) - L_f^{(r)} h(x(t)), \qquad \forall\ t \in [t_0, t_0+\varepsilon)$$

and hence,

$$y^{(r)}(t) = y_d^{(r)}(t), \qquad \forall\ t \in [t_0, t_0+\varepsilon)$$

and using the hypothesis of the lemma that

$$y_d^{(k)}(t_0) = L_f^k h(x_0) \qquad \forall\, k = 0, 1, \ldots, r-1$$

repeated integration yields

$$\Gamma_{x_0}^O(u(t)) = y_d(t), \qquad \forall\ t \in [t_0, t_0 + \varepsilon)$$

which is the required result. □

The result in the previous lemma is easy to comprehend by looking at the expressions for the output derivatives (3.2). As control $u(t)$ does not directly affect $y^{(k)}(t)$, $\forall k = 1, \cdots, r-1$, their values at t_0 are determined by the initial state. The control u, for which $\Gamma_{x_0}^O(u) = y_d(\cdot)$, is given by (3.3) with y replaced by y_d in that formula.

We can now easily check for the switch-singular pairs among Γ_p, Γ_q with relative degrees r_p, r_q respectively, where p, $q \in \mathcal{P}$.

Lemma 3 *For SISO switched systems, (x_0, y) is a switch-singular pair of two subsystems Γ_p and Γ_q if and only if $y \in \mathcal{Y}_p \cap \mathcal{Y}_q$ and*

$$\begin{pmatrix} y \\ \vdots \\ y^{(r_\kappa - 1)} \end{pmatrix} (t_0) = \begin{pmatrix} h_\kappa(x_0) \\ \vdots \\ L_{f_\kappa}^{r_\kappa - 1} h_\kappa(x_0) \end{pmatrix}, \quad \kappa = p, q \in \mathcal{P}. \tag{3.8}$$

The lemma basically states that if there are two subsystems satisfying Equation (3.5) for an output y and initial state x_0, then that output is indistinguishable between these two subsystems, and such (x_0, y) forms a switch-singular pair.

The example below illustrates the use of these concepts.

Example 2 Consider a SISO switched system with two modes

$$\Gamma_p := \begin{cases} \dot{x} = \begin{pmatrix} x_1 + x_2 \\ x_2 \\ x_1 x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ x_2 \end{pmatrix} u, \quad \mathbb{M} = \mathbb{R}^3, \\ y = x_1 \end{cases}$$

$$\Gamma_q := \begin{cases} \dot{x} = \begin{pmatrix} x_2 \\ x_2 x_3 \\ -x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ x_2 \end{pmatrix} u, \quad \mathbb{M} = \mathbb{R}^3. \\ y = 2x_1 \end{cases}$$

If Γ_p is active, then $\dot{y} = x_1 + x_2$; if Γ_q is active, then $\dot{y} = 2x_2$. Both subsystems have relative degree 2 on $\mathbb{R}^3$ which can be verified by taking the second derivative of the output. If there exists $x \in \mathbb{R}^3$ which forms a switch-singular pair with $y \in \mathcal{Y}_p \cap \mathcal{Y}_q$, then the following equality must be satisfied:

$$\begin{pmatrix} x_1 \\ x_1 + x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$$

which gives $x_1 = x_2 = 0$. This state constraint yields $y = \dot{y} = 0$. If we let $\mathcal{Y}^\alpha := \left\{ y(t) \mid \dot{y}(t) \in \mathcal{F}^{AC} \text{ and } \begin{pmatrix} y(t) \\ \dot{y}(t) \end{pmatrix} \neq 0 \ \forall t \right\}$, then there exists no switch-singular pair between $x \in \mathbb{R}^3$ and $y \in \mathcal{Y}^\alpha$. Theorem 1 and Lemma 1 infer that the switched system generated by $\{\Gamma_p, \Gamma_q\}$ is strongly invertible on $\mathcal{Y}^\alpha$, $\forall x_0 \in \mathbb{R}^3$. $\lhd$

We now have the tool set to check for the invertibility conditions given in Theorem 1. If these conditions are satisfied and the switched system is strongly invertible, a switched inverse system can be constructed to recover the input u and the switching signal σ from a given output and initial state. For the switched inverse system, define the *index inversion function* $\overline{\Sigma}^{-1} : \mathbb{M}^{\alpha} \times \mathcal{Y}^{\alpha} \to \mathcal{P}$ as

$$\overline{\Sigma}^{-1}(x_0, y) = p \ : \ y \in \mathcal{Y}_p \text{ and } y^{(k)}(t_0) = L_{f_p}^{k} h_p(x_0) \tag{3.9}$$

where $k = 0, 1, \cdots, r_p - 1$, t_0 is the initial time of y, and $x_0 = x(t_0)$. The function $\overline{\Sigma}^{-1}$ is well-defined since p is unique by the fact that there are no switch-singular pairs. The existence of p is guaranteed because it is assumed that $y \in \mathcal{Y}^{\alpha}$ is an output. Thus, an inverse switched system Γ_{σ}^{-1} is

$$\begin{aligned}
\sigma(t) &= \overline{\Sigma}^{-1}(z(t), y(t)), && \text{(3.10a)}\\
\dot{z}(t) &= f_{\sigma}(z) + g_{\sigma}(z)\big(- \frac{L_{f_\sigma}^{r_\sigma} h_{\sigma}(z)}{L_{g_\sigma} L_{f_\sigma}^{r_\sigma - 1} h_{\sigma}(z)} + \frac{y^{(r_\sigma)}(t)}{L_{g_\sigma} L_{f_\sigma}^{r_\sigma - 1} h_{\sigma}(z)} \big), && \text{(3.10b)}\\
u(t) &= - \frac{L_{f_\sigma}^{r_\sigma} h_{\sigma}(z)}{L_{g_\sigma} L_{f_\sigma}^{r_\sigma - 1} h_{\sigma}(z)} + \frac{y^{(r_\sigma)}(t)}{L_{g_\sigma} L_{f_\sigma}^{r_\sigma - 1} h_{\sigma}(z)} && \text{(3.10c)}
\end{aligned}$$

with the initial condition $z(t_0) = x_0$. The notation $(\cdot)_{\sigma}$ denotes the object calculated for the subsystem with index $\sigma(t)$. The initial condition $\sigma(t_0)$ determines the initial active subsystem at the initial time t_0, from which time onwards, the active subsystem indexes and the input as well as the state are determined uniquely and simultaneously via (3.10).

3.1.1 SISO bilinear systems

In general, it is computationally hard to solve Equation (3.8) for switch-singular pairs. In case of SISO switched bilinear systems, the nonlinear equations in (3.8) become linear. This makes the task of checking the existence of switch-singular pairs between two subsystems comparatively easier, which can now be achieved by comparing the ranks of certain matrices. To this end, consider a switched system which comprises SISO bilinear subsystems having the dynamics of the form

$$\begin{aligned}\dot{x} &= A_{\sigma(t)}x + B_{\sigma(t)}xu,\\ y &= C_{\sigma(t)}x\end{aligned} \tag{3.11}$$

where $\sigma(t) = p \in \mathcal{P}$, $x \in \mathbb{R}^n$, $A_p, B_p \in \mathbb{R}^{n\times n}$, $C_p \in \mathbb{R}^{1\times n}$. Also, $u(t),\ y(t) \in \mathbb{R}$.

If some mode $p \in \mathcal{P}$ is active over a time interval, then at any time t in that interval, the expression for derivatives of output is

$$\begin{aligned}y(t) &= C_p x(t),\\ \dot{y}(t) &= C_p A_p x(t),\\ &\vdots\\ y^{(r_p-1)}(t) &= C_p A_p^{r_p-1} x(t),\\ y^{(r_p)}(t) &= C_p A_p^{r_p} x(t) + C_p A_p^{r_p-1} B_p x u(t)\end{aligned} \tag{3.12}$$

where r_p denotes the relative degree of subsystem p. If we introduce the notations

$$\hat{y}_p(t) := \begin{pmatrix} y(t) \\ \dot{y}(t) \\ \vdots \\ y^{(r_p-1)(t)} \end{pmatrix} \qquad \text{and} \qquad Z_p := \begin{pmatrix} C_p \\ C_p A_p \\ \vdots \\ C_p A_p^{r_p-1} \end{pmatrix},$$

then based on the functional reproducibility criteria, an output $y(t)$ can be produced by a subsystem p if and only if $\hat{y}_p(t) = Z_p x(t)$. Consequently, if two subsystems p, q

can produce a given segment of output on an interval $[t_0, t_0+\varepsilon)$, then we will have

$$\begin{pmatrix} \hat{y}_p(t_0) \\ \hat{y}_q(t_0) \end{pmatrix} = \begin{pmatrix} Z_p \\ Z_q \end{pmatrix} x(t_0)\ . \tag{3.13}$$

This is equivalent to saying that

$$\left[\begin{array}{c} I_p \\ \hline I_q \end{array}\right] \begin{pmatrix} y(t_0) \\ \dot{y}(t_0) \\ \vdots \\ y^{(r-1)}(t_0) \end{pmatrix} = \left(\begin{array}{c} Z_p \\ \hline Z_q \end{array}\right) x(t_0) \tag{3.14}$$

where $r = \max\{r_p, r_q\}$, and for $\kappa = \{p, q\}$, I_κ is an $r_\kappa \times r$ matrix whose ij^{th} element is 1 if $i = j$ and 0 otherwise. Thus, the existence of switch-singular pairs in case of SISO bilinear switched systems implies that the intersection of range spaces of $\begin{pmatrix} I_p \\ I_q \end{pmatrix}$ and $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$ is not empty. Since $\begin{pmatrix} I_p \\ I_q \end{pmatrix}$ and $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$ are both linear operators acting on linear subspaces, the *zero vector* is always in their range space. Thus, an identically zero output always forms a switch singular pair with the kernel of $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$; that is, $(\ker \begin{pmatrix} Z_p \\ Z_q \end{pmatrix}, 0)$ forms the switch-singular pair for such systems. That is the trivial case; for the nontrivial case we check if $\begin{pmatrix} I_p \\ I_q \end{pmatrix}$ and $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$ have a nontrivial common range space. In other words, if there exists a nonzero output that forms a switch-singular pair with some state at time t, then $\begin{pmatrix} y(t) \\ \dot{y}(t) \\ \vdots \\ y^{(r-1)}(t) \end{pmatrix} \in$ range $\begin{pmatrix} I_p \\ I_q \end{pmatrix} \cap$ range $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$ or equivalently rank $\begin{bmatrix} I_p & Z_p \\ I_q & Z_q \end{bmatrix} <$ rank $\begin{bmatrix} I_p \\ I_q \end{bmatrix}$+rank $\begin{bmatrix} Z_p \\ Z_q \end{bmatrix}$. This discussion leads to the following result.

Proposition 1 *Consider a single-input single-output switched bilinear system (3.11) with all subsystems being invertible. Let* $r = \max_{p \in \mathcal{P}} p$*. For all* $x(t_0) := x_0 \in \mathbb{R}^n$ *and* $y \in \mathcal{Y}^\alpha := \{y \mid y^{(r-1)} \in \mathcal{F}^{AC} \text{ and } y_{[t,t+\varepsilon)} \not\equiv 0,\ \forall\, t \geq t_0,\ \forall\, \varepsilon > 0\}$*, the pairs* (x_0, y) *are not switch-singular pairs of* Γ_p, Γ_q*, if and only if the following rank condition holds:*

$$rank \begin{bmatrix} I_p & Z_p \\ I_q & Z_q \end{bmatrix} = rank \begin{bmatrix} I_p \\ I_q \end{bmatrix} + rank \begin{bmatrix} Z_p \\ Z_q \end{bmatrix} \tag{3.15}$$

for all $p \neq q,\ p, q \in \mathcal{P}$ *such that* $\mathcal{Y}_p \cap \mathcal{Y}_q \neq \{0\}$.

Proof. We already developed the proof of the necessity part. For the proof of the sufficiency part, assume that the rank condition (3.15) holds; then range $\begin{pmatrix} I_p \\ I_q \end{pmatrix} \cap$ range $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix} = \{0\}$. If there exists $(x_0, y(t))$ that forms a switch-singular pair, that is, $\begin{pmatrix} \hat{y}_p(t_0) \\ \hat{y}_q(t_0) \end{pmatrix} = \begin{pmatrix} Z_p \\ Z_q \end{pmatrix} x(t_0)$, then Equation (3.14) must hold. Since the common range space only consists of $\{0\}$, (3.14) holds true only for $y(t) \equiv 0$ so that only the zero output forms a switch-singular pair with the kernel of $\begin{pmatrix} Z_p \\ Z_q \end{pmatrix}$. Such

$y(t) \notin \mathcal{Y}^\alpha$, implying that there exist no switch-singular pairs, and that proves the sufficiency part. $\square$

The result given in Proposition 1 is exactly the same as that given in [9, Lemma 3], where the authors gave conditions for checking the existence of switch-singular pairs in switched linear systems. The common framework in both cases is as follows: when one writes the expression for derivatives of the outputs, one gets linear equations on the right-hand side. So, for checking the existence of switch-singular pairs, one has to work with linear equations, which makes it easier to derive the rank conditions. In general, such equations are not linear. To see this, consider the system given in (3.1). For simplicity, assume that $\mathcal{P} = \{p, q\}$ and the subsystems Γ_p, Γ_q have equal relative degrees, i.e., $r_p = r_q =: r$. Lemma 3 states that Γ_p, Γ_q have a switch-singular pair (x_0, y) if and only if

$$\hat{y} = \mathcal{H}_p(x_0) = \mathcal{H}_q(x_0) \tag{3.16}$$

where $\hat{y} = (y, \dot{y}, \cdots, y^{(r-1)})^T$ and $\mathcal{H}_\kappa = (h_p, L_{f_p}h_p, \cdots, L_{f_p}^{r-1}h_p)^T$, $\kappa = \{p, q\}$. To see if there exist any switch-singular pairs, one is interested in solving (3.16) for x_0; that is, switch-singular pairs actually lie in the solution space of r-nonlinear equations where each equation itself involves functions of an n-dimensional variable x. As it is hard to talk to about the solutions of nonlinear equations in general, investigation into more constructive conditions for checking of switch-singular pairs is a topic of ongoing research.

3.2 Multiple-Input Multiple-Output (MIMO) Systems

In this section, we seek a result similar to Lemma 3 for multi-input multi-output switched nonlinear systems. For multiple-input multiple-output (MIMO) non-switched nonlinear systems affine in controls (2.1), one uses the *structure algorithm* to compute the inverse. When a system is invertible, the structure algorithm, or Singh's inversion algorithm, allows us to express the input as a function of the output, its derivatives and possibly some states.

The Structure Algorithm: This version of the algorithm closely follows the construction given in [17], which is a slightly modified version of the algorithm in [13].

Step 1: Calculate

$$\dot{y} = L_f h(x) + L_G h(x)u = \frac{\partial h}{\partial x}[f(x) + G(x)u]$$

and write it as $\dot{y} =: a_1(x) + b_1(x)u$. Define $s_1 := \operatorname{rank} b_1(x)$, which is the maximal rank of $b_1(x)$ in some neighborhood of x_0, denoted as $N_1(x_0)$. Permute, if necessary, the components of the output so that the first s_1 rows of $b_1(x)$ are linearly dependent. Decompose y as

$$\dot{y} = \begin{pmatrix} \dot{\tilde{y}}_1 \\ \dot{\hat{y}}_1 \end{pmatrix} = \begin{pmatrix} \tilde{a}_1(x) + \tilde{b}_1(x)u \\ \hat{a}_1(x) + \hat{b}_1(x)u \end{pmatrix}$$

where $\dot{\tilde{y}}_1$ consists of the first s_1 rows of $\dot{y}$. Since the last $m - s_1$ rows of $b_1(x)$ are linearly dependent upon the first s_1 rows, there exists a matrix $F_1(x)$ such that

$$\begin{aligned} \dot{\tilde{y}}_1 &= \tilde{a}_1(x) + \tilde{b}_1(x)u, \\ \dot{\hat{y}}_1 &= \hat{h}^1(x, \dot{\tilde{y}}_1) = \hat{a}_1(x) + F_1(x)(\dot{\tilde{y}}_1 - \tilde{a}_1(x)) \end{aligned} \tag{3.17}$$

where the last equation is affine in $\dot{\tilde{y}}_1$. Finally, set $\tilde{B}_1(x) := \tilde{b}_1(x)$.

Step k+1: Suppose that in steps 1 through k, $\dot{\tilde{y}}_1, \ldots, \tilde{y}_k^{(k)}$, $\hat{y}_k^{(k)}$ have been defined so that

$$
\begin{aligned}
\dot{\tilde{y}}_1 &= \tilde{a}_1(x) + \tilde{b}_1(x)u, \\
&\vdots \\
\tilde{y}_k^{(k)} &= \tilde{a}_k(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k-1, i \le j \le k\}) \\
&\quad + \tilde{b}_k(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k-1, i \le j \le k-1\})u, \\
\hat{y}_k^{(k)} &= \hat{h}^k(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k\})
\end{aligned}
$$

where all the expressions on the right-hand side are rational functions of $\tilde{y}_i^{(j)}$. Suppose also that the matrix $\tilde{B}_k := [\tilde{b}_1^T, \ldots, \tilde{b}_k^T]^T$ (vertical stacking of the linearly independent rows obtained at each step) has full rank equal to s_k in $N_k(x_0)$. Then calculate

$$
\hat{y}_k^{(k+1)} = \frac{\partial}{\partial x}\hat{h}^k[f(x) + G(x)u] + \sum_{i=1}^{k}\sum_{j=i}^{k} \frac{\partial \hat{h}^k}{\partial \tilde{y}_i^{(j)}} \tilde{y}_i^{(j+1)}
$$

and write it as

$$
\begin{aligned}
\hat{y}_k^{(k+1)} &= a_{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k+1\}) \\
&\quad + b_{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k\})u \; .
\end{aligned} \tag{3.18}
$$

Define $B_{k+1} := [\tilde{B}_k^T, \, b_{k+1}^T]^T$, and $s_{k+1} := \operatorname{rank} B_{k+1}$. Permute, if necessary, the components of $\hat{y}_k^{(k+1)}$ so that the first s_{k+1} rows of B_{k+1} are linearly independent. Decompose $\hat{y}_k^{(k+1)}$ as

$$
\hat{y}_k^{(k+1)} = \begin{pmatrix} \tilde{y}_{k+1}^{(k+1)} \\ \\ \hat{y}_{k+1}^{(k+1)} \end{pmatrix}
$$

where $\tilde{y}_{k+1}^{(k+1)}$ consists of the first $(s_{k+1} - s_k)$ rows. Since the last $(m - s_{k+1})$ rows of

$B_{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k\})$ are linearly dependent on the first s_{k+1} rows, we can write

$$
\begin{aligned}
&\dot{\tilde{y}}_1 = \tilde{a}_1(x) + \tilde{b}_1(x)u, \\
&\vdots \\
&\tilde{y}_{k+1}^{(k+1)} = \tilde{a}_{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k+1\}) \\
&\qquad + \tilde{b}_{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k, i \le j \le k\})u, \\
&\hat{y}_{k+1}^{(k+1)} = \hat{h}^{k+1}(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le k+1, i \le j \le k+1\})
\end{aligned}
$$

where once again everything is rational in $\tilde{y}_i^{(j)}$. Finally, set $\tilde{B}_{k+1} := [\tilde{B}_k^T, \tilde{b}_{k+1}^T]^T$, which has full rank equal to s_{k+1} locally.

End of Step $k+1$.

By construction, $s_1 \le s_2 \le \cdots \le m$. If for some integer α we have $s_\alpha = m$, then the algorithm terminates. We call α the *relative order*[1] of the system. If $s_n < m$, then such an α does not exist. The closed form expression for u is derived from the α-th step of the structure algorithm which yields

$$
\begin{aligned}
&\dot{\tilde{y}}_1 = \tilde{a}_1(x) + \tilde{b}_1(x)u, \\
&\vdots \\
&\tilde{y}_\alpha^{(\alpha)} = \tilde{a}_\alpha(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le \alpha - 1, i \le j \le \alpha\}) \\
&\qquad + \tilde{b}_\alpha(x, \{\tilde{y}_i^{(j)} | \, 1 \le i \le \alpha - 1, i \le j \le \alpha - 1\})u \, .
\end{aligned}
\tag{3.19}
$$

So, one can construct an invertible matrix $\tilde{B}_\alpha := [\tilde{b}_1^T, \ldots, \tilde{b}_\alpha^T]^T$ having full rank equal

[1]The term was coined in [12] and is weaker than the notion of vector relative degree.

to m in a neighborhood $N_\alpha(x_0) =: N(x_0)$.

$$u(t) = \tilde{B}_\alpha^{-1} \left[\begin{pmatrix} \dot{\tilde{y}}_1 \\ \vdots \\ \dot{\tilde{y}}_\alpha \end{pmatrix} - \begin{pmatrix} \tilde{a}_1 \\ \vdots \\ \tilde{a}_\alpha \end{pmatrix} \right] =: \tilde{B}_\alpha^{-1}[\tilde{Y}_\alpha - \tilde{A}_\alpha] \cdot \tag{3.20}$$

Note that the entries of the matrix $\tilde{B}_\alpha$ are rational functions of the derivatives of the output and there may exist an output for which the rank of $\tilde{B}_\alpha$ drops. All such outputs are called *singular outputs* and we define $\mathcal{Y}_p^s := \{y \in \mathcal{Y}_p \,|\, \operatorname{rank} \tilde{B}_\alpha(x,y) < m, x \in N(x_0)\}$. Hence, we work with u such that $\Gamma_{x_0}^O(u) \notin \mathcal{Y}_p^s$ for any time instant. Comparing to the SISO case, we had $\tilde{B}_\alpha = L_g L_f^{r-1} h(x)$ which is a function of the state only, and thus there exists no singular output for SISO systems. Another useful class of systems for which $\mathcal{Y}_p^s = \emptyset$ was discussed in [12] by Hirschorn, as the matrix $\tilde{B}_\alpha$ for such systems is independent of output and its derivatives. As was the case in SISO systems, substitution of the expression for u from (3.20) in (2.1) gives the dynamics of the inverse system. These dynamics are defined on an open and dense set $\mathbb{M}^\alpha := \{x \in \mathbb{M} \,|\, \operatorname{rank} \tilde{B}_\alpha(x,y) = m, y \notin \mathcal{Y}_p^s\}$.

However, unlike in the SISO case, we need some differentiability assumptions on the input signals to characterize the input space for MIMO systems. In the structure algorithm, Step 1 gives $\dot{\tilde{y}}_1$ that already has u on the right-hand side, and the α-th step of the algorithm involves $\{\tilde{y}_i^{(j)} \,|\, 1 \le i \le \alpha - 1, i \le j \le \alpha\}$. Thus $\tilde{y}_i^{(\alpha-1)}$ must be absolutely continuous so that $\tilde{y}_i^{(\alpha)}$ exists almost everywhere. For the input space, it means that $u^{(\alpha-1)}$ must be Lebesgue measurable and locally essentially bounded. These constraints characterize the input space $\mathcal{U}$ for MIMO case.

Based on the structure algorithm, we now study the conditions for functional reproducibility of multivariable nonlinear systems. Using the notation derived in the structure algorithm, denote by Z the vector

$$Z\Big(x, \dot{\tilde{y}}_1, \cdots, \tilde{y}_{\alpha-1}^{(\alpha-1)}\Big) := \begin{pmatrix} h(x) \\ \hat{h}^1(x, \dot{\tilde{y}}_1) \\ \vdots \\ \hat{h}^{\alpha-1}\Big(x, \dot{\tilde{y}}_1, \cdots, \tilde{y}_{\alpha-1}^{(\alpha-1)}\Big) \end{pmatrix} \tag{3.21}$$

and let

$$\hat{y} := \begin{pmatrix} y \\ \hat{y}_1 \\ \vdots \\ \hat{y}_{\alpha-1}^{(\alpha-1)} \end{pmatrix}, \qquad \hat{y}_d := \begin{pmatrix} y_d \\ \hat{y}_{d_1} \\ \vdots \\ \hat{y}_{d_{\alpha-1}}^{(\alpha-1)} \end{pmatrix} \cdot \tag{3.22}$$

So Z is basically a concatenation of the expressions appearing at each step of Singh's structure algorithm which get differentiated and $\hat{y}$ is the concatenation of the corresponding expressions on the left-hand side so that

$$Z\Big(x, \dot{\tilde{y}}_1, \cdots, \tilde{y}_{\alpha-1}^{(\alpha-1)}\Big) - \hat{y} = 0 \cdot$$

The following result is along the same line as Lemma 2.

Lemma 4 *If the system given by Equation (2.1) is strongly invertible at $x(t_0) = x_0$ and has a relative order $\alpha < \infty$, then there exists $\varepsilon > 0$ and a control input u over the interval $[t_0, t_0 + \varepsilon)$ such that $\Gamma_{x_0}^{O}(u(t)) = y_d(t)$ for all $t \in [t_0, t_0 + \varepsilon)$ if and only if*

$$\hat{y}_d(t_0) = Z\big(x_0, \dot{\tilde{y}}_{d_1}(t_0), \cdots, \tilde{y}_{d_k}^{(k)}(t_0)\big) \quad \forall k = 0, 1, \cdots, \alpha - 1 \tag{3.23}$$

where $\hat{y}_d$ is defined as in (3.22).

Proof *Necessity:* Supposing $\exists\, \varepsilon > 0$ and input u defined over the interval $[t_0, t_0+\varepsilon)$, such that $\Gamma^O_{x_0}(u(t)) = y_d(t)$, $\forall t \in [t_0, t_0 + \varepsilon)$, then

$$\begin{aligned}
y_d(t_0) &= y(t_0) = h(x_0) \\
\hat{y}_{d_1}(t_0) &= \hat{y}_1(t_0) = \hat{h}^1(x, \dot{\tilde{y}}_1) = \hat{h}^1(x, \dot{\tilde{y}}_{d_1}) \\
&\ \ \vdots \\
\hat{y}^{(\alpha-1)}_{d_{\alpha}-1}(t_0) &= \hat{y}^{(\alpha-1)}_{\alpha-1}(t_0) = \hat{h}^{\alpha-1}\Big(x_0, \dot{\tilde{y}}_1, \cdots, \tilde{y}^{(\alpha-1)}_1, \cdots, \tilde{y}^{(\alpha-1)}_{\alpha-1}\Big) \\
&= \hat{h}^{\alpha-1}\Big(x_0, \dot{\tilde{y}}_{d_1}, \cdots, \tilde{y}^{(\alpha-1)}_{d_1}, \cdots, \tilde{y}^{(\alpha-1)}_{d_{\alpha-1}}\Big)
\end{aligned} \tag{3.24}$$

and hence Equation (3.23) is satisfied.

Sufficiency: If we inject $y_d(t)$ into the inverse system, then the control input produced by this inverse system is given by (3.20) with $\tilde{y}$ replaced by $\tilde{y}_d$, and substituting it in (3.19) gives

$$\dot{\tilde{y}}_1(t) = \dot{\tilde{y}}_{d_1}(t), \qquad \forall\, t \in [t_0, t_0 + \varepsilon)\ \cdot \tag{3.25}$$

Here $t_0+\varepsilon$ characterizes the time instant at which the trajectory of the inverse system hits the singular point in the state space. As the system is strongly invertible at x_0, it is guaranteed that $\varepsilon > 0$.

Using hypothesis (3.23), we have $h(x_0) = y_d(t_0)$, and integrating (3.25) on both sides over the interval $[t_0, t_0 + \varepsilon)$ to get

$$\tilde{y}_1(t) = \tilde{y}_{d_1}(t), \qquad \forall\, t \in [t_0, t_0 + \varepsilon)\ \cdot \tag{3.26}$$

Using the initial conditions characterized by (3.23), the desired result can now be derived by induction. Suppose Equations (3.25) and (3.26) are true for index k, that is

$$\begin{aligned}
\tilde{y}_k^{(k)}(t) &= \tilde{y}_{d_k}^{(k)}(t) \qquad \forall\, t \in [t_0, t_0+\varepsilon), \\
\tilde{y}_k(t) &= \tilde{y}_{d_k}(t) \qquad \forall\, t \in [t_0, t_0+\varepsilon)\,.
\end{aligned}$$

Since $\tilde{y}_{k+1}^{(k+1)} = \tilde{a}_{k+1}(x, \{\tilde{y}_i^{(j)} \,|\, 1 \le i \le k, i \le j \le k+1\}) + \tilde{b}_{k+1}(x, \{\tilde{y}_i^{(j)} \,|\, 1 \le i \le k, i \le j \le k\})u$, substituting u from (3.20) as generated by the inverse system, with $\tilde{y}$ replaced by $\tilde{y}_d$, gives

$$\tilde{y}_{k+1}^{(k+1)}(t) = \tilde{y}_{d_{k+1}}^{(k+1)}(t) \qquad \forall\, t \in [t_0, t_0+\varepsilon)\,.$$

Again using hypothesis (3.23) and integrating both sides, we get

$$\tilde{y}_{k+1}(t) = \tilde{y}_{d_{k+1}}(t) \qquad \forall\, t \in [t_0, t_0+\varepsilon)\,.$$

As $y(t) = (\tilde{y}_1(t), \ldots, \tilde{y}_\alpha(t))$, we get $\Gamma_{x_0}^O(u(t)) = y_d(t)$, $\forall\, t \in [t_0, t_0+\varepsilon)$. $\square$

An insightful geometric version of this result in terms of jet spaces is given in [18]. Similarly to the SISO case, the idea is that the portion of output which is not directly affected by u is determined initially by the value of state variables; and the input u, for which $\Gamma_{x_0}^O(u) = y_d(\cdot)$, is given by (3.20) with y replaced by y_d in that formula.

Example 3 Consider the system given in Example 1. The vector $\hat{y}$ is the portion of the output that gets differentiated, and therefore

$$\hat{y} = \begin{pmatrix} y_1 \\ y_2 \\ \dot{y}_2 \end{pmatrix} \quad \Rightarrow \quad \hat{y}_d = \begin{pmatrix} y_{d_1} \\ y_{d_2} \\ \dot{y}_{d_2} \end{pmatrix}$$

and the vector $Z(x, y_1, y_2, \dot{y}_{d_1})$ is given by

$$Z(x, y_1, y_2, \dot{y}_{d_1}) = \begin{pmatrix} x_1 \\ x_2 \\ \dot{y}_{d_1}(x_3/x_1) \end{pmatrix} .$$

By Lemma 4 and Equation (2.2), if we have $\hat{y}_d(t_0) = Z\big((x_0, y_1(t_0), y_2(t_0), \dot{y}_{d_1}(t_0)\big)$, then the control which produces y_d as an output, on a small interval, is given by

$$\begin{aligned} u_1 &= \frac{\dot{y}_{d_1}}{x_1} \\ u_2 &= \frac{x_1\ddot{y}_{d_2} - x_3\ddot{y}_{d_1} + \dot{y}_{d_1}\dot{y}_{d_2}}{\dot{y}_{d_1}} . \end{aligned}$$

If $y_d(t) \notin \mathcal{Y}^s$ for all time instants and the corresponding state trajectory $x(t) \in \mathbb{M}^\alpha$, then the system can produce y_d as an output over arbitrary time interval. $\lhd$

The result of Lemma 4 gives the following condition for the verification of switch-singular pairs.

Lemma 5 *For MIMO switched systems, (x_0, y) is a switch-singular pair of two subsystems Γ_p, Γ_q if and only if $y \in \mathcal{Y}_p \cap \mathcal{Y}_q$ and*

$$\begin{pmatrix} y \\ \dot{\hat{y}}_1 \\ \vdots \\ \hat{y}^{\alpha_\kappa - 1}_{(\alpha_\kappa - 1)} \end{pmatrix} = \begin{pmatrix} h_\kappa(x_0) \\ \hat{h}^1_\kappa(x_0, \dot{\tilde{y}}_1) \\ \vdots \\ \hat{h}^{\alpha_\kappa - 1}_\kappa(x_0, \dot{\tilde{y}}_1, \cdots, \tilde{y}^{(\alpha_\kappa - 1)}_{\alpha_\kappa - 1}) \end{pmatrix} \tag{3.27}$$

where α_κ denotes the relative order of subsystems Γ_κ and $\kappa = p, q$.

As was the case in SISO systems, the lemma just states that if a given output $y(t)$ and an initial state x_0 satisfy Equation (3.23) for any two MIMO subsystems, then such (x_0, y) form switch-singular pairs.

Thus, we have developed the tools for checking the invertibility of subsystems and the existence of switched-singular pairs. Based on the result in Theorem 1, one can verify these two characteristics to infer the invertibility of MIMO switched nonlinear systems. To reconstruct the input and the switching signal, the procedure for constructing the inverse system from this point onwards is exactly the same as discussed earlier for the SISO case.

We now provide an example as an illustration of Lemma 5.

Example 4 Consider a switched system with $\mathcal{P} = \{p, q\}$, $u = (u_1, u_2)^T$, $y = (y_1, y_2)^T$ and $\mathbb{M} = \mathbb{R}^3$. The dynamics of the subsystems are as follows:

$$\Gamma_p := \begin{cases} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} u_1 \\ x_3 u_1 \\ u_2 \end{pmatrix}, \\ \\ \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \end{cases} \qquad \Gamma_q := \begin{cases} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} x_2^3 \\ u_1 \\ u_2 \end{pmatrix} \\ \\ \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_3 \end{pmatrix} \end{cases}$$

If at any time instant, Γ_p is active, then the expression for derivatives of output are

$$\begin{aligned} &\dot{y}_1 = u_1 \\ &\dot{y}_2 = x_3 u_1 = x_3 \dot{y}_1 \quad \Rightarrow \quad \ddot{y}_2 = x_3 \ddot{y}_1 + \dot{y}_1 u_2 \ . \end{aligned}$$

So, $\alpha_p = 2$ and Γ_p is invertible as long as $\dot{y}_1 \neq 0$. If $x(t_0) := x_0 = (x_{10}, x_{20}, x_{30})^T$, $\sigma_{[t_0, t_0+\varepsilon)} = p$, then under the assumption that right-derivatives are well defined, the output must satisfy

$$\begin{pmatrix} y_1(t_0) \\ y_2(t_0) \\ \dot{y}_2(t_0) \end{pmatrix} = \begin{pmatrix} x_{10} \\ x_{20} \\ x_{30}\dot{y}_1(t_0) \end{pmatrix} \cdot \tag{3.28}$$

On the other hand, if Γ_q is active at any time instant, then the expression for derivatives of output are

$$\begin{aligned} &\dot{y}_1 = x_2^3 \quad \Rightarrow \quad \ddot{y}_1 = 3x_2^2 u_1 \\ &\dot{y}_2 = u_2 \cdot \end{aligned}$$

Again, $\alpha_q = 2$ and Γ_q is invertible as long as $x_{20} \neq 0$ and if $\sigma_{[t_0, t_0+\varepsilon)} = q$, the output must satisfy

$$\begin{pmatrix} y_1(t_0) \\ y_2(t_0) \\ \dot{y}_1(t_0) \end{pmatrix} = \begin{pmatrix} x_{10} \\ x_{30} \\ x_{20}^3 \end{pmatrix} \cdot \tag{3.29}$$

According to Lemma 5, if there exists a switch-singular pair (x_0, y), then this pair must satisfy (3.28) and (3.29). So, every switch singular pair is characterized by the following constraints:

$$x_{10} = y_1(t_0), \quad x_{20} = x_{30} = y_2(t_0) = \frac{\dot{y}_2(t_0)}{\dot{y}_1(t_0)}, \quad \dot{y}_1(t_0) = x_{20}^3 \quad \text{with} \quad x_{20}, \dot{y}_1(t_0) \neq 0 \cdot$$

Equivalently, every initial condition of the form $(a, b, b)^T$ forms a switch-singular pair with $\{y_{[t_0, t_0+\varepsilon)} : y_1(t_0) = a, y_2(t_0) = b, \dot{y}_1(t_0) = b^3, \dot{y}_2(t_0) = b^4\}$. $\triangleleft$

We conclude this chapter by providing a few remarks which extend the applicability of our results.

Remark 3 *Our results can be extended to include the more general case of different* *output dimensions. For switched systems whose subsystems are of different output dimensions, the formulae of switch-singular pairs remain unchanged. Lemma 5 implies that the output dimensions of the two subsystems must be the same in order for* (x_0, y) *to be a singular pair. If the output dimensions are different, then it is automatically true that there is no singular pair for the two systems since* $\mathcal{Y}_p \cap \mathcal{Y}_q = \emptyset$*. If subsystem output dimensions are different, what is needed is the use of "hybrid functions" to describe the output set* $\mathcal{Y}$ *of the switched system (which are now not functions but concatenations of functions of different dimensions), but other than that, the statements of Theorem 1 and Corollary 1 remain the same.*

Remark 4 *The results in this chapter can also be extended to include the case when there are state jumps at switching times. Denote by* $\psi_{p,q} : \mathbb{M} \to \mathbb{M}$ *the* reset map *when switching from subsystem p to subsystem q, p, q* $\in \mathcal{P}$*, that is, if* τ *is a switching time,*

$$x(\tau) = x(\tau^+) = \psi_{\sigma(\tau^-),\sigma(\tau)}(x(\tau^-)) \cdot$$

Thus far, we have considered the case of identity reset maps only, where $\psi_{p,q}(x) = x$ $\forall\, p, q \in \mathcal{P}, \forall x \in \mathbb{M}$*. For nonidentity reset maps, Definition 3 is modified to "*(x_0, y) *is a switch-singular pair of the two subsystems* Γ_p*,* Γ_q *if there exist* u_1*,* u_2 *and* $\varepsilon > 0$ *such that* $\Gamma^O_{p,x_0}(u_{1[t_0,t_0+\varepsilon)}) = \Gamma^O_{q,\psi_{p,q}(x_0)}(u_{2[t_0,t_0+\varepsilon)}) = y_{[t_0,t_0+\varepsilon)}$ *or* $\Gamma^O_{p,\psi_{q,p}(x_0)}(u_{1[t_0,t_0+\varepsilon)}) = \Gamma^O_{q,x_0}(u_{2[t_0,t_0+\varepsilon)}) = y_{[t_0,t_0+\varepsilon)}$*." Essentially, this means that the output is indistinguishable between the two subsystems, taking into account the effect of state jumps. Equation (3.8) becomes*

$$\begin{pmatrix} y \\ \vdots \\ y^{(r_\kappa - 1)} \end{pmatrix}(t_0) = \begin{pmatrix} h_\kappa(\psi_{p,\kappa}(x_0)) \\ \vdots \\ L^{r_\kappa - 1}_{f_\kappa} h_\kappa(\psi_{p,\kappa}(x_0)) \end{pmatrix}, \qquad \kappa = p, q \in \mathcal{P},$$

or

$$\begin{pmatrix} y \\ \vdots \\ y^{(r_\kappa-1)} \end{pmatrix}(t_0) = \begin{pmatrix} h_\kappa(\psi_{q,\kappa}(x_0)) \\ \vdots \\ L_{f_\kappa}^{r_\kappa-1} h_\kappa(\psi_{q,\kappa}(x_0)) \end{pmatrix}, \qquad \kappa = p, q \in \mathcal{P}$$

where $\psi_{p,p}(x_0) = \psi_{q,q}(x_0) = x_0$, $\forall\ p, q \in \mathcal{P}$. *Equation (3.27) would also be modified similarly when dealing with MIMO systems. The statement of Theorem 1, in either case, remains unchanged.*

Another generalization is to include switching mechanisms, such as switching surfaces. *Denote by* $S_{p,q}$ *the switching surface for system* p, *where the switched system jumps to subsystem* q,

$$x(t) = \psi_{p,q}(x(t^-)) \qquad \text{if } x(t^-) \in S_{p,q} \quad \text{and} \quad \sigma(t^-) = p.$$

Then we only need to check for the switch-singularity of $x_0 \in S_{p,q}$ *and* $x_0 \in S_{q,p}$ *instead of* $x_0 \in \mathbb{M}$ *for the two subsystems* Γ_p, Γ_q.

CHAPTER 4

OUTPUT TRACKING

In the previous chapters, we considered the question of left invertibility where the objective was to recover $(\sigma,\ u)$ uniquely for all y in some output set $\mathcal{Y}^{\alpha}$. In this section, we address a different problem which is concerned with finding $(\sigma,\ u)$ that may not be unique, such that $\mathrm{H}_{x_0}(\sigma,\ u) = y_d$ for a given function y_d and a state x_0. For the invertibility problem, we found conditions on the subsystems and the output set $\mathcal{Y}$ so that the map H_{x_0} is injective for all x_0 in some subset. Here, we are given one particular $(x_0,\ y_d)$ and wish to find its preimage under the map H_{x_0}. For the switched system (2.3), denote by $\mathrm{H}_{x_0}^{-1}$ the preimage of a function y_d,

$$\mathrm{H}_{x_0}^{-1} := \{(\sigma,\ u)\ :\ \mathrm{H}_{x_0}(\sigma,\ u) = y_d\}\cdot \tag{4.1}$$

If y_d is not in the image set of H_{x_0} then by convention $\mathrm{H}_{x_0}^{-1} = \emptyset$. When $\mathrm{H}_{x_0}^{-1}(y_d)$ is a singleton, the map H_{x_0} is invertible at y. We want to find conditions and an algorithm to generate $\mathrm{H}_{x_0}^{-1}(y_d)$ when $\mathrm{H}_{x_0}^{-1}(y_d)$ is a finite set.

We require the individual subsystems to be strongly invertible because if this is not the case, then the set $\mathrm{H}_{x_0}^{-1}(y_d)$ may be infinite. For a noninvertible nonswitched nonlinear system,[1] the matrix $\tilde{B}_{\alpha}^{-1}$ in (3.20) is not defined and the expression for u is modified to

$$u(t) = \tilde{B}_{\alpha}^{\dagger}[\tilde{Y}_{\alpha} - \tilde{A}_{\alpha}] + K(x, \tilde{Y}_{\alpha-1})v \tag{4.2}$$

[1] A nonswitched system is not invertible if it has more inputs than outputs or it does not satisfy the structure algorithm criteria.

where K is a matrix whose columns form a basis for the null space of $\tilde{B}_\alpha$ and $\tilde{B}_\alpha^\dagger := \tilde{B}_\alpha^T(\tilde{B}_\alpha \tilde{B}_\alpha^T)^{-1}$ is a right pseudo-inverse of $\tilde{B}_\alpha$. If an output is generated by some input u obtained from (4.2) with some initial state, then due to arbitrary choice of v, there always exist infinitely many different inputs that generate the same output with the same initial state. Hence to avoid infinite loop reasoning, we will assume that the individual subsystems Γ_p are strongly invertible for all $p \in \mathcal{P}$. However, we do not assume that the switched system is invertible as the subsystems may have switch-singular pairs. We will only consider the functions $y_d(t)$ over finite time intervals so that there is only a finite number of switches under consideration.

Casting a quick glance at the problem, it seems that one solution is straightforward: switchings occur at times when the output loses continuity, and since we have finite numbers of systems and switches, there is a finite number of possible switching signals and an exhaustive search would suffice. However, this is far from being the only case because we can have a switching and the output is still smooth at that switching time. As we shall see, it is possible to use a switch at a later time to recover a "hidden switch" earlier (e.g., a switch at which the output is smooth) and this phenomenon is peculiar to switched systems with no counterpart in non-switched systems. Also, when dealing with nonlinear dynamics, it is entirely possible that the desired output trajectory $y_{d[t_0,T]}$ could only be produced by a single subsystem Γ_p at time t_0, but later on y_d passes through the set $\mathcal{Y}_p^s$ (the set of singular outputs for Γ_p) at some time $\tau < T$. In that case, one must switch to another subsystem $\{\Gamma_q | y_d(\tau) \notin \mathcal{Y}_q^s\}$, at time instant τ, to have any possibility of recovering the preimage of $y_{d[\tau,T]}$. These issues make the recovery of the input and the switching signal more complex for switched nonlinear systems.

We now present a switching inversion algorithm for switched systems similar to the one given in [9]. The algorithm actually uses the *semigroup property* for the trajectories of dynamical systems from which it follows that $\mathrm{H}_{x_0}(\sigma_{[t_0,T)}, u_{[t_0,T)})(t) = \mathrm{H}_{x(s)}(\sigma_{[s,T)}, u_{[s,T)})$ where $x(s) = \Gamma_{x_0,\sigma_{[t_0,s)}}(u_{[t_0,s)})$, for any $s \in [t_0,\, T)$ and any $t \geq s$. Based on that, the algorithm determines the preimage on a subinterval $[t_0,\, t)$ of $[t_0,\, T)$ and then concatenates these with the corresponding preimage on the rest of the interval $[t,\, T)$.

The algorithm takes the parameters $x_0 \in \mathbb{M}$, $y_d \in \mathcal{F}^{pc}$ (defined over a finite interval) and returns the set $\mathrm{H}_{x_0}^{-1}(y_d)$. It uses the *index-matching map*[2] $\Sigma^{-1} : \mathbb{M} \times \mathcal{F}^{pc} \to 2^{\mathcal{P}}$ defined as $\Sigma^{-1}(x_0,\, y_d) := p$ such that $y_d \in \mathcal{Y}_p^{\alpha}$ and y_d satisfies (3.23), obtained via the structure algorithm of Γ_p. The index-matching map returns the indexes of the subsystems that are capable of generating y_d starting from x_0. If the returned set is empty, no subsystem is able to generate that y_d starting from x_0. In the algorithm, $\Gamma_{p,x_0}^{-1,O}(y_d)$ denotes the output of the inverse subsystem Γ_p^{-1}; the symbol "$\leftarrow$" reads "assigned as", and ":=" reads "defined as". Recall that the concatenation of an element η and a set S is $\eta \oplus S := \{\eta \oplus \zeta, \zeta \in S\}$. By convention, $\eta \oplus \emptyset = \emptyset$, $\forall \eta$. Finally, the concatenation of two sets S and T is $S \oplus T := \{\eta \oplus \zeta,\, \eta \in S,\, \zeta \in T\}$.

Note that the index-matching map Σ^{-1} is defined for every pair $(x_0,\, y_d)$ and always returns a set, whereas the index inversion function $\overline{\Sigma}^{-1}$ (3.9) is defined only for $(x_0,\, y_d)$, which are not switch-singular pairs, and returns an element of $\mathcal{P}$.

If the return is a nonempty set, the set must be finite and contains pairs of switching signals and inputs that generate the given y_d starting from x_0. If the return is an empty set, it means that there is no switching signal and input that generate y_d, or there is an infinite number of possible switching times. Also by our concatenation notation: if at any instant of time, the return of the procedure is an empty set, then that branch of the search will be empty because $\eta \oplus \emptyset = \emptyset$.

[2]The set $2^{\mathcal{P}}$ denotes the set of all subsets of the set $\mathcal{P}$.

begin $\mathrm{H}_{x_0}^{-1}(y_d)$

Let the domain of y_d be $[t_0, T)$.

Let $\overline{\mathcal{P}} := \{p \in \mathcal{P} : y_{d[t_0,t_0+\varepsilon)} \in \mathcal{Y}_p^\alpha$ and $x_0 \in \mathbb{M}_p^\alpha,\ \varepsilon > 0\}$

Let $t^* := \min\{t \in [t_0, T) : y_{d[t,t+\varepsilon)} \notin \mathcal{Y}_p^\alpha$ for some $p \in \overline{\mathcal{P}},\ \varepsilon > 0\}$ otherwise $t^* = T$.

Let $\mathcal{P}^* := \Sigma^{-1}(x_0,\ y_{d[t_0,t_0+\varepsilon)})$.

if $\mathcal{P}^* \neq \emptyset$ **then**

Let $\mathcal{A} := \emptyset$

foreach $p \in \mathcal{P}^*$ **do**

Let $x := \Gamma_{p,x_0}^{-1}(y_{d[t_0,t^*)})$

if $x \in \mathbb{M}_p^\alpha$ *and* $y_{d[t_0,t^*)} \in \mathcal{Y}_p^\alpha$ **then**

Let $u := \Gamma_{p,x_0}^{-1,O}(y_{d[t_0,t^*)})$

$\mathcal{T} := \{t \in (t_0, t^*) : (x(t), y_d(t))$ is a switch- singular pair of Γ_p, Γ_q

for some $q \neq p\}$.

if $\mathcal{T}$ *is a finite set* **then**

foreach $\tau \in \mathcal{T}$ **do**

let $\xi := \Gamma_p(u)(\tau)$

$\mathcal{A} \leftarrow \mathcal{A} \cup \{(\sigma_{[t_0,\tau)},\ u_{[t_0,\tau)}) \oplus \mathrm{H}_\xi^{-1}(y_{d[\tau,T)})\}$

else if $\mathcal{T} = \emptyset$ *and* $t^* < T$ **then**

let $\xi := \Gamma_p(u)(t^*)$

$\mathcal{A} \leftarrow \mathcal{A} \cup \{(\sigma_{[t_0,t^*)},\ u) \oplus \mathrm{H}_\xi^{-1}(y_{d[t^*,T)})\}$

else if $\mathcal{T} = \emptyset$ *and* $t^* = T$ **then**

$\mathcal{A} \leftarrow \mathcal{A} \cup \{(\sigma_{[t_0,T)}, u)\}$

else

$\mathcal{A} := \emptyset$

else

$\mathcal{A} := \emptyset$

else

$\mathcal{A} := \emptyset$

return $\mathrm{H}_{x_0}^{-1}(y_d) := \mathcal{A}$

end

As stated earlier, the algorithm basically determines the preimage on a subinterval $[t_0, t)$ of $[t_0, T)$ and then concatenates these with the corresponding pre-image on the rest of the interval $[t, T)$. If t is the first switching time after t_0, then we can find $H_{x_0}^{-1}(y_{d[t_0,t)})$ by singling out which subsystems are capable of generating $y_{d[t_0,t)}$ using the index-matching map. The obvious candidate for first switching time, denoted by t^* in the algorithm, is the time at which the output loses smoothness. Note that in the SISO case, t^* is the time at which one of the first $r-1$ derivatives of the output loses continuity (see Section 3.1). But it is entirely possible that we have a switching and the output is still smooth at that switching time because of a switch-singular pair (see Example 5). The algorithm takes that into account and uses a switch at a later time to recover a "hidden switch" earlier (e.g., a switch at which the output is smooth). This makes the switching inversion algorithm a recursive procedure calling itself with different parameters within the main algorithm (e.g., the function $\mathrm{H}_{x_0}^{-1}(y_d)$ uses the returns of $\mathrm{H}_{\xi}^{-1}(y_{d[t^*,T)})$).

The following example will help understand this algorithm.

Example 5 Consider a switched system with two modes

$$\Gamma_1 : \begin{cases} \dot{x} = \begin{pmatrix} x_1 x_2 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u, \quad \mathbb{M} = \mathbb{R}^2 \\ y = x_2 \end{cases}$$

$$\Gamma_2 : \begin{cases} \dot{x} = \begin{pmatrix} 0 \\ x_1 \end{pmatrix} + \begin{pmatrix} e^{x_2} \\ e^{x_2} \end{pmatrix} u, \quad \mathbb{M} = \mathbb{R}^2 \cdot \\ y = x_1 \end{cases}$$

We wish to reconstruct the switching signal $\sigma(t)$ and the input $u(t)$ which will generate the following output:

$$y_d(t) = \begin{cases} \cos t & \textit{if } t \in [0,\, t^*) \\ 2\cos t & \textit{if } t \in [t^*,\, T) \end{cases}$$

where $t^* = \pi$ and $T = 4.5$, with the given initial state $x_0 = (0,\, 1)^T$.

In this example, any state x lying on the diagonal $\Delta := \{(x_1, x_2)^T : x_1 = x_2\}$ forms a switch-singular pair with the output whose corresponding state trajectory hits the same state x at any time.

We now use the above switching inversion algorithm to find $(\sigma,\, u)$ such that $\Gamma^O_{x_0,\sigma}(u) = y_d$. We have $\mathcal{P}^* := \Sigma^{-1}(x_0,\, y_{d[0,t^*)}) = \{1\}$ by using the index-matching map with given x_0 and $y_d(0) = 1$. The inverse of Γ_1 on $[0,\, t^*)$ is

$$\Gamma_1^{-1} : \begin{cases} \dot{z} = \begin{pmatrix} z_1 z_2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \dot{y}_d, \quad \mathbb{M}_1^\alpha = \mathbb{R}^2 \\ u(t) = -z_2 + \dot{y}_d \end{cases}$$

with $z(0) = x_0$, which then gives

$$\begin{aligned} z(t) &= \begin{pmatrix} 0 \\ \cos t \end{pmatrix} =: x(t) \\ u(t) &= -\cos t - \sin t \end{aligned} \qquad t \in [0, t^*). \tag{4.3}$$

We want to find $\mathcal{T} := \{t \le t^* : (x(t), y_{d[t,t^*)})$ is a switch-singular pair of Γ_1, Γ_2 $\}$, which is equivalent to solving

$$\cos t = x_1(t) = 0, \quad t \in (0,\, t^*).$$

This equation has a solution $t = \pi/2 =: t_1 < t^*$, and hence $\mathcal{T} = \{t_1\}$, a finite set. We have $\xi = x(t_1) = (0,0)^T$ and we repeat the procedure for the initial state ξ and the output $y_{d[t_1,T)}$ with $\mathcal{P}^* := \Sigma^{-1}(\xi,\, y_{d[t_1,t^*)}) = \{1, 2\}$. We analyze these two cases:

Case 1 $p = 1$. This implies t_1 is not a switching time and $u(t)$, $x(t)$ are still given by (4.3) for $t_1 \le t < t^*$. Repeating the procedure with $\xi = x(t^*) = (0,\, 0)^T$ and $y_{d[t^*,T)}$ and $y_d(t^*) = -2$, we observe that $y_d(t^*) \neq x_1(t^*)$ and also $y_d(t^*) \neq x_2(t^*)$; thus the index-matching map returns an empty set, $\Sigma^{-1}(\xi, y_{d[t^*,T)}) = \emptyset$.

Case 2 $p = 2$, which means that t_1 is a switching instant. So we work with the inverse system of Γ_2,

$$\Gamma_2^{-1} : \begin{cases} \dot{z} = \begin{pmatrix} 0 \\ z_1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \dot{y}_d, \quad \mathbb{M}_1^{\alpha} = \mathbb{R}^2 \\ u(t) = e^{-z_2}\, \dot{y}_d \end{cases}$$

with initial state $z(t_1) = \xi$, which gives

$$\begin{aligned} z(t) &= \begin{pmatrix} \cos t \\ \cos t + \sin t - 1 \end{pmatrix} =: x(t) \\ u(t) &= -e^{\cos t + \sin t}\ \sin t \end{aligned} \qquad t \ge t_1.$$

We find $\mathcal{T} = \{t_1 < t \le t^* : (x(t), y_{d[t,t^*)})$ is a switch-singular pair of $\Gamma_1,\, \Gamma_2\}$, which is equivalent to solving for

$$\cos t = x_2(t) = \cos t + \sin t - 1, \qquad \frac{\pi}{2} = t_1 < t \le t^* = \pi\,\cdot$$

It is easy to see that this equation has no solution and thus there exist no switch-singular pairs in interval $(t_1,\, t^*)$. So, we let $\xi = x(t^*) = (-1,\, -2)^T$ and repeat the procedure with ξ and $y_{d[t^*,T)}$, which gives the unique solution $\sigma_{[t^*,T)} = 1$ and $u_{[t^*,T)} = -2(\cos t + \sin t)$.

Thus, the switching inversion algorithm returns $(\sigma,\, u)$, where

$$\sigma(t) = \sigma^1_{[0,t_1)} \oplus \sigma^2_{[t_1,t^*)} \oplus \sigma^1_{[t^*,T)}$$

$$u(t) = \begin{cases} -\cos t - \sin t, & \text{if } 0 \le t < t_1 \\ -e^{\cos t + \sin t}\ \sin t, & \text{if } t_1 \le t < t^* \\ -2(\cos t + \sin t), & \text{if } t^* \le t \le T\,. \end{cases}$$

Here, the switching signal and the input, $(\sigma,\, u)$, recovered by the switching inversion algorithm happen to be the unique one. $\triangleleft$

In the above example, the output only loses smoothness at t^* and t^* is a switching instant. However, there is another switching at t_1 where the output does not lose smoothness. Without the concept of switch-singular pairs, one might falsely conclude that there is no switching signal and input that generates $y_d(t)$, but instead the use of the switching inversion algorithm allows us to recover the input and switching signal.

CHAPTER 5

APPLICATIONS

Invertibility of switched systems basically refers to the ability to identify the mode of operation and reconstruct the continuous inputs using the knowledge of the measured output. Since various engineering systems can be modeled as switched systems, invertibility of switched systems has several possible applications. For example, a multimodal aircraft can be modeled as a switched system where the switching signals indicate which mode the aircraft is currently in. The invertibility property relates to the problem of detecting the flying mode of the aircraft using the measured information, which could be the path and/or the velocity of the aircraft on ground radar equipment. We now discuss two applications in detail.

5.1 Fault Detection

The issue of invertibility is an interesting open problem in the context of hybrid systems, whose significance can be recognized in failure detection application (for example, communication, automotive and power systems) or in safety critical applications (for example, air traffic systems). Given a switched system, the idea is to mark some modes (discrete states) as "critical" that correspond to an unsafe behavior or to a fault whereas the other modes represent nominal operation of the system. Using invertibility, we investigate if it is possible to detect when a system enters a critical state. As a by product, one may reconstruct the input that led the system into the critical/faulty mode.

In this section, we apply our invertibility results to a single machine connected to an infinite bus. A classical model is used to describe the dynamics of a single-machine connected to an infinite bus [26, Chapter 5]. The system may operate in two different modes. The first corresponds to the operation under nominal condition and the other corresponds to the case where one of the lines opens as a result of fault. Given the observation of a state variable, the problem is to detect which line failed open.

The system under consideration is a single-machine connected to an infinite bus with voltage V_∞ through two parallel lines with complex impedances $j2X_l$ and jX_l, respectively, as depicted in Fig. 5.1. X_m denotes the machine inductance and E_t is the internal voltage of the machine. As explained in [26], the dynamics of this kind of system can be described independently of the type of machine model by $\dot{x} = f(x, V_\infty, u)$, where x denotes the states of the machine, and u is the machine control input. In this case, the single-machine is the system of interest and the dynamics take the form $\dot{x} = (x, V_\infty, u) = Ax + \psi(x, V_\infty) + Ku$, which is also the general structure of multimachine dynamic models.

For the single-machine model shown in Fig. 5.1, let $x = (\delta, \omega_r)^T$ denote the state of the machine. Here, δ denotes the angular position of the rotor in electrical radians, and ω_r denotes the angular velocity of the rotor in rad/s measured with respect to stationary references δ_0 and ω_s respectively. In the case of the classical model, the dynamics take the following simplified form:

$$\Gamma_0 : \begin{cases} \begin{pmatrix} \dot{\delta} \\ \dot{\omega}_r \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -D\frac{\omega_s}{2H} \end{pmatrix} \begin{pmatrix} \delta \\ \omega_r \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{\omega_s}{2H}\frac{3E_t}{3X_m+2X_l}\sin\delta \end{pmatrix} V_\infty + \begin{pmatrix} 0 \\ \frac{\omega_s}{2H} \end{pmatrix} T_m \\ y = \omega_r \end{cases} \tag{5.1}$$

where D is the per unit damping torque coefficient, H is the per unit inertia constant, and T_m is the mechanical torque. All the parameters in (5.1) are assumed to be constant.

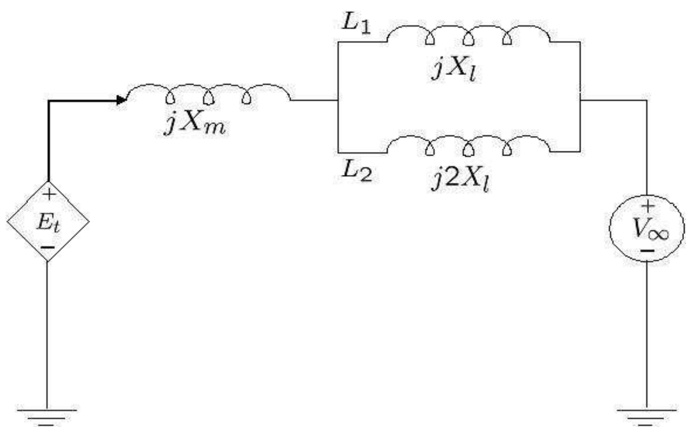

Figure 5.1: Synchronous machine classical model dynamic circuit

A fault in line L_1 will cause the line to open, thus the system dynamics will be modified as follows:

$$\Gamma_1 : \begin{cases} \begin{pmatrix} \dot{\delta} \\ \dot{\omega}_r \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -D\frac{\omega_s}{2H} \end{pmatrix} \begin{pmatrix} \delta \\ \omega_r \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{\omega_s}{2H}\frac{E_t}{X_m+2X_l}\sin\delta \end{pmatrix} V_\infty + \begin{pmatrix} 0 \\ \frac{\omega_s}{2H} \end{pmatrix} T_m. \\ y = \omega_r \end{cases} \tag{5.2}$$

Similarly, a fault in line L_2 results in the following modified dynamics:

$$\Gamma_2 : \begin{cases} \begin{pmatrix} \dot{\delta} \\ \dot{\omega}_r \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -D\frac{\omega_s}{2H} \end{pmatrix} \begin{pmatrix} \delta \\ \omega_r \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{\omega_s}{2H}\frac{E_t}{X_m+X_l}\sin\delta \end{pmatrix} V_\infty + \begin{pmatrix} 0 \\ \frac{\omega_s}{2H} \end{pmatrix} T_m. \\ y = \omega_r \end{cases} \tag{5.3}$$

Given the observation of ω_r, we want to find out if any of the lines failed open, and if so, which ones. Viewing $\Gamma := \{\Gamma_0, \Gamma_1, \Gamma_2\}$ as a switched system, the problem is to identify the mode of operation and detect the change in mode of operation.

As Γ_0 is active under nominal conditions, it is assumed that Γ_0 is a stable system and has achieved its steady state value $(\delta_0, \omega_{r,0})^T$. Furthermore, we assume that V_∞ and T_m (the external inputs) also remain constant when Γ_0 is in steady state. Our invertibility results do not apply here directly as the inputs remain constant; this means that even though V_∞ and T_m appear in the expression of $\dot{y} = \dot{\omega}_r$, it does not affect y in an arbitrary manner. Due to this, we have to analyze the output equation and the system dynamics a bit more closely, but the concepts are still the same.

If there is no fault in the system, then output $y(t)$ satisfies following equations:

$$\begin{aligned} y(t) &= \omega_r(t) && \text{(5.4a)} \\ \dot{y}(t) &= \dot{\omega}_r(t) = -D\frac{\omega_s}{2H}\omega_r(t) - \frac{\omega_s}{2H}\frac{E_t}{X_m + 2X_l}\sin\delta\, V_\infty + \frac{\omega_s}{2H}T_m\,. && \text{(5.4b)} \end{aligned}$$

Now, suppose that a fault has occurred in L_1 at time τ_1, then Γ_1 is active and the output $y(\tau_1)$ satisfies the following two equations:

$$\begin{aligned} y(\tau_1) &= \omega_r(\tau_1) && \text{(5.5a)} \\ \dot{y}(\tau_1) &= \dot{\omega}_r(\tau_1) = -D\frac{\omega_s}{2H}\omega_r(\tau_1) - \frac{\omega_s}{2H}\frac{E_t}{X_m + 2X_l}\sin\delta\, V_\infty + \frac{\omega_s}{2H}T_m. && \text{(5.5b)} \end{aligned}$$

Similarly, if a fault occurs in L_2 at time instant τ_2, then Γ_2 is active and the output $y(\tau_2)$ satisfies the following two equations:

$$\begin{aligned} y(\tau_2) &= \omega_r(\tau_2) && \text{(5.6a)} \\ \dot{y}(\tau_2) &= \dot{\omega}_r(\tau_2) = -D\frac{\omega_s}{2H}\omega_r(\tau_2) - \frac{\omega_s}{2H}\frac{E_t}{X_m + 2X_l}\sin\delta\, V_\infty + \frac{\omega_s}{2H}T_m. && \text{(5.6b)} \end{aligned}$$

Equation (5.4a) is the same as (5.5a) and (5.6a), so one cannot tell anything about the mode of operation by looking at $y(t)$ without having a look at its derivatives. The expressions for $\dot{y}$ corresponding to Γ_0, Γ_1 and Γ_2, given by (5.4b), (5.5b) and

(5.6b) respectively, contain more information about the dynamics of the subsystems and provide a test for fault detection. So at any time t, if $\dot{y}(t)$ satisfies *only* (5.4b), then one can conclude that the system is operating without any fault. Otherwise, if $\dot{y}(t)$ satisfies (5.5b), respectively (5.6b), we conclude that there is a fault in L_1, respectively L_2. The only matter of concern is whether there exists a pair $(x(t), y(t))$ which satisfy any two equations among (5.4b), (5.5b) and (5.6b) simultaneously. Such pairs are actually the switch-singular pairs and play the role of an obstacle in fault detection; that is, one cannot detect the fault in the system if the state and the output of the system form a switch-singular pair. To compute the switch-singular pairs, we solve (5.4b) = (5.5b) for $(\delta, \omega_r)^T$; using simple arithmetic, we see that it is possible if and only if $\delta = \{0, \pi, 2\pi, \cdots\}$ or $X_l = 0$ (no inductance in lines) or $X_l = \infty$ (both lines fail open simultaneously). Solving (5.4b) = (5.6b), and (5.5b) = (5.6b) yields similar conditions. From system specifications, $X_l \neq 0$ and δ cannot be an integral multiple of π. So, unless both lines fail open simultaneously, we can always detect the time and type of fault that occurs.

5.2 Topology Recovery in Multiagent Networks

Multiagent systems have appeared broadly in several applications including formation flight of unmanned air vehicles (UAVs), clusters of satellites, self-organization, automated highway systems, and congestion control in communication networks (see [27, 28] and references therein). An important problem that appears frequently in the context of coordination of multiagent systems is the group agreement or consensus problem [28, 29, 30].

A topology of a multiagent network is a graph G whose nodes represent agents and edges between nodes denote information flow between the corresponding agents.

For a node i, denote by $\mathcal{N}_i$ the set of neighbors of node i, where neighbors of a node i are all other nodes that have an edge with node i. The dynamics of agent i are denoted by $\dot{x}_i = f_i(x_i, u_i)$, where x_i is the state and u_i is the control input. In cooperative multiagent systems, the agents' control law can use not only its own state but also information from neighboring agents, and the control law is of the form $u_i = g_i(x_i, \{x_j, j \in \mathcal{N}_i\})$. The collective dynamics of the network of agents are obtained by putting together all the agent dynamics and take the form $\dot{x} = F_G(x)$, where $x = (x_1, \cdots, x_n)$ and F_G is a function depending on the structure of the graph G. For example (see [28]), with the agent dynamics $\dot{x}_i = u_i$ and the cooperative control law $u_i = \sum_{j\in\mathcal{N}_i}(x_j - x_i)$ (known as *linear consensus protocol*), the collective network dynamics are $\dot{x} = -L(G)x$, where $L(G) = [l_{ij}]$, $l_{ij} = \begin{cases} -1 & \text{if } j \in \mathcal{N}_i, j \neq i \\ |\mathcal{N}_i| & \text{if } i = j \end{cases}$ is the graph Laplacian of G. Hence, F_G takes the linear form because of linear control law.

But more often than not, the dynamic agents represent some physical models, so one has to take actuator saturation into account. This naturally leads us to the design and analysis of *nonlinear consensus protocols*. For example, with integrator nodes $\dot{x}_i = u_i$, every node in the network attains a common decision value with the nonlinear consensus protocol $u_i = \sum_{j\in\mathcal{N}_i}\phi_{ij}(x_j - x_i)$, where $\phi_{ij} = \phi_{ji}$ is continuous and locally Lipschitz, $\phi_{ij}(0) = 0$, $\phi_{ij}(-x) = -\phi_{ij}(x)\ \forall x \in \mathbb{R}$, $(x-y)(\phi_{ij}(x) - \phi_{ij}(y)) > 0\ \forall x \neq y$. An example of such a function would be $\phi(x) = x^3$; or $\phi(y) = \arctan(y)$ if one is particularly interested in finding a bounded function. The details on convergence of nonlinear consensus protocols are given in [29]. The collective network dynamics on applying this control law are given by $\dot{x} = F_G(x) = (f_1, f_2, \cdots, f_n)^T$, where $f_i = \sum_{j\in\mathcal{N}_i}\phi_{ij}(x_j - x_i)$.

Moreover, if the network under consideration consists of mobile agents, then it is natural to assume that some of the existing communication links may fail due to an obstacle between two agents or new links may be established over the period of time as the change in position of agents would affect the range of detection between one another. In terms of network topology, this means that a certain number of edges are added or removed from the graph G, which implies that the network topology varies over time. Since the number of nodes is fixed and finite, G can take only finitely many values, that is $G : [0, T] \to \{G_p, p \in \mathcal{P}\}$. We then have a network with switching topology. Networks with switching topology can be modeled as switched systems where the subsystems are the network dynamics with fixed topologies and the switching signal, in this case, indicates the active topology at every time. In particular, the switched system describing multiagent networks with switching topology is $\dot{x} = F_\sigma(x)$, where $\dot{x} = F_p(x)$ is the collective dynamics of the network with topology G_p, $p \in \mathcal{P}$, and $\sigma(t) := p : G(t) = G_p$. Observe that the closed loop dynamics of each agent depend upon its neighboring agents; change in network topology, therefore, changes the dynamics of each agent.

We now investigate the invertibility properties of a network with switching topology where each node has nonlinear dynamics. Identification of discrete state in this case determines the topology of the network. If disturbances are to be incorporated, the switched system under consideration is given by

$$\dot{x} = F_\sigma(x) + d \tag{5.7}$$

where $d : [0, T] \to \mathbb{R}^n$. Assume that there is an output y coming out of the system and $h_p(x)$ is the observation associated with topology G_p, $p \in \mathcal{P}$; then

$$y = h_\sigma(x) \tag{5.8}$$

is the output equation.

If (5.7) and (5.8) form an invertible switched system, then reconstruction of $\sigma(t)$ over an interval $[0, T]$ reveals the active topology at each time instant in that interval.

For a concrete example, consider an undirected network with three nodes which can switch between two topologies $G_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $G_2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. If c_{ij} denotes the ij^{th} element of matrix G_κ, $\kappa = \{1, 2\}$, then $c_{ij} = 1$ indicates a communication link between node i and node j; similarly, $c_{ij} = 0$ shows that there exists no link between node i and node j, where $i, j \in \{1, 2, 3\}$. If the output with topology G_1 is $y = \begin{pmatrix} e^{x_2}e^{-x_1} \\ x_1 e^{x_2} \\ x_3 \end{pmatrix}$ and the output with topology G_2 is $y = \begin{pmatrix} e^{-x_1^3}e^{x_2} \\ x_3 e^{x_2} \\ 2x_3 \end{pmatrix}$, then, using Equation (3.27), we infer that there are switch-singular pairs. To be precise, Equation (3.27) yields that every initial state of the form $(0, a, 0)^T$, $a \in \mathbb{R}$, forms a switch-singular pair with the output whose initial value is $(e^a, 0, 0)^T$, and hence, one cannot recover the network topology for an arbitrary state and an arbitrary output. On the other hand, if the output with topology G_1 is $y = \begin{pmatrix} e^{-x_2^2}e^{x_1} \\ x_1 e^{x_2} \\ x_3 \end{pmatrix}$ and the output with topology G_2 is $y = \begin{pmatrix} 2e^{x_1} \\ x_3 e^{x_2} \\ 2x_3 \end{pmatrix}$, there exist no switch-singular pairs and we can guarantee topology recovery, in the presence of disturbance, for all output y.

Apart from the two applications mentioned above, invertibility may serve as a useful structural property in *communication networks*, where the system under consideration is a network that consists of a number of nodes and there are data packets flowing along communication links between nodes. To avoid congestion, network traffic is regulated by transmission control protocols (TCPs), which tell nodes to drop packets if there is excessive incoming data compared to outgoing data. When dropping occurs, dynamics of the nodes change abruptly and this fact leads to the use of hybrid systems to model communication networks. Invertibility may represent a desired property in system design, because it allows one to infer the system configuration (characterized by the operating mode) and the external influences (characterized by inputs) from the observed data. Or, to ensure system security, one may want to design the system not to be invertible, in order to ensure that the intruder will not be able to gain access to protected system information.

To sum up, we conclude this chapter by saying that the issue of invertibility is an interesting open problem in the context of hybrid systems, whose significance can be recognized in various engineering applications.

CHAPTER 6

CONCLUSIONS

In the thesis, we addressed an unexplored problem, namely, the invertibility of switched nonlinear systems. This problem seeks a condition on a switched system that guarantees unique reconstruction of the switching signal and the input from an output and a given initial state. A necessary and sufficient condition for the invertibility of switched systems was given which required the invertibility of subsystems and the nonexistence of switch-singular pairs. The concept of switch-singular pairs introduced in [9] for linear systems was extended to nonlinear systems. We then developed the formula for checking if (x_0, y) is a switch-singular pair of two subsystems. In the case of single-input single-output bilinear systems, the formula for checking the existence of a switch-singular pair reduces to checking the rank of some matrices. Assuming that all the subsystems are invertible, we develop an algorithm to recover the input and switching signal that generate a desired output trajectory from given initial state. In the end, we highlighted some of the potential application scenarios.

For future work, one interesting problem is to develop conditions for checking the existence of switch-singular pairs which are more constructive as it is in general not feasible to verify (3.27) for every output and state. Another research direction is to approach the problem geometrically and investigate characterizations equivalent to nonexistence of switch-singular pairs to obtain geometric criteria for left invertibility of switched systems. The geometric criteria for invertibility, even for switched linear systems, may provide more constructive conditions for system design because it does not require us to look at the derivatives of the output. In doing so, one may look for similarities among the subsystems which exhibit switch-singular pairs.

It would also be interesting to look at the problem of output generation when the subsystems are not invertible. This relaxation allows us to cover a larger class of switched nonlinear systems, but the analysis requires us to study the reachability sets of nonlinear systems.

By and large, invertibility is an important property in system design and system security analysis. The theoretical tools developed in the thesis will provide new guidelines for system design in applications. We envision applications in a variety of other areas, including vehicle systems, biological systems, and power networks.

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